

Unified Accelerator Fabric (UAF)

Engineering Specification for Implementation (UAF-SPEC-001)

UAF Working Group

Version 2.1.4 — Draft for Approval

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Document Control

Version	Date	Author	Changes
0.9	—	UAF WG	Initial draft
1.0	2026-01-01	UAF WG	Full engineering specification, PDF-ready
2.0	2026-09-29	UAF WG	CXL/NVMe doorbell ordering; userspace dma-heap + FastRPC SMMU mapping; C ABI completed; 32-bit rkey; 16-byte packed DST CQE
2.1.4	2026-09-29	UAF WG	Errata over v2.1.3. The CXL window bounds check is stated in an overflow-safe form, because the sum in <code>remote_addr + L > cxl_size</code> wraps for <code>uint64_t</code> ; [R-9.5-004] no longer cites a packet-counting throttle on a path with no packets; the optional data-packet authenticator of [R-11.2-008], which had no byte layout, is withdrawn; [R-9.5-006] no longer contradicts [R-9.5-009] about where the completion ring is; the Section 6.7 command-specific fallback row is realigned; scope and sample-payload version strings updated

Version	Date	Author	Changes
2.1.3	2026-09-29	UAF WG	Closes the review of v2.1.2: the path byte is on the connection record, so the discriminant survives the only byte exchange the document defines; [R-5.5-012], cited by the conformance suite and never written, now exists, and the requirement index fails the build on any such dangling citation; the intra-host completion inherits the initial phase of 1 and the wrap flip; the NVMe status codes are corrected against Base 2.0d; the ring offset has a formula; stale cross-references repaired
2.1.2	2026-09-29	UAF WG	Closes the review of v2.1.1: a path discriminant with a third RTR column for the CXL window; the CM authenticator defined for a 32-byte body and replay keyed on the header qp_id; the endpoint identity filled from source fields; a layout for the intra-host RMT completion; per-source rkey throttling where no link authenticator is configured; rkey bound to a queue pair at connect; stale references to UAF_OP_NAK, UAF_QPT_UD and UAF_CAP_MT_POST removed; version strings reconciled at 2.1.2

Version	Date	Author	Changes
2.1.1	2026-09-29	UAF WG	Closes the review of v2.1: key rules scoped to the software table so UAF-N is satisfiable; a public storage completion carrying the full 64-bit <code>wr_id</code> ; a symmetric simultaneous-open tie-break; NAK subtypes, an ECN-echo bit and an ACK schedule; the rkey failure limit no longer changes queue-pair state; the missing host structs and flag enumerations defined; atomic completion semantics moved to Section 4 so Profile A is covered; the UAF-D intra-host binding, ring consumer and NVMe status map specified; corrected MAP_SYNC rationale, consumer-side acquire, and two arithmetic errors
2.1	2026-09-29	UAF WG	Closes both v2.0 reviews. One data plane per backend and two declared wire profiles; message framing, atomic responses, CM packet layouts; defined poll return convention; <code>uaf_query_device</code> ; in-place registration; one CSPRNG key algorithm; NVMe mapping replaces the unused pseudo-SQE; corrected reliability and performance figures; RFC 2119 language with numbered requirements and an executable conformance suite

Classification: Engineering Internal

Status: Draft for Approval. This revision is **not** approved for implementation. Approval re-

quires the sign-off block in Section 1.6 to be completed and the open issues in Appendix D to be closed or accepted.

Target platforms: NVIDIA (H100/B200/GB200), Qualcomm Snapdragon (X Elite, 8 Gen 3-4), Qualcomm Dragonfly (C1000).

Breaking changes from v2.0

v2.0 was never implemented, so these changes are made cleanly rather than carried as compatibility debt. Every one of them closes a defect that made the prior text unimplementable or ambiguous.

Change	Reason
uaf_poll_cq / uaf_storage_poll return a count, not UAF_OK	The convention was undefined and the two candidate readings are incompatible once $\max > 1$
UAF_ERR_CQ_EMPTY (-8) retired and its value reserved	An empty queue is 0 entries, not an error; no other error code is renumbered
Wire header re-laid out; lkey_rsvd removed, msg_id, msg_len, seg_off, data_crc32c added	A multi-packet message could not be framed and the payload was outside every integrity check
UAF_WIRE_MTU_DEFAULT now counts the 64-byte header	v2.0's maximum packet was 4160 bytes and did not fit a 4 KiB path MTU
UAF_WR_RECV removed from the wire opcode table	It is a local buffer post, not a packet
No enum or size_t field in any ABI struct; every struct pinned by _Static_assert	enum width is implementation-defined; size_t differs between ILP32 and LP64 Snapdragon userspace
uaf_mr.pal_priv removed from the public struct	It froze every backend's internals into the application ABI
dma_buf_fd removed from uaf_conn_info	A file descriptor cannot be carried in a connection exchange
uaf_storage_cqe no longer packed; gains a phase tag	packed dropped alignment to 1; without a phase tag a zeroed slot is indistinguishable from UAF_OK
uaf_storage_close(sq, cr) takes both objects	v2.0 closed only the submission queue
Section 6.1's 64-byte "UAF SQE" deleted	No backend ever filled it; DST is specified as an NVMe mapping
uaf_create_qp takes struct uaf_qp_init_attr;	Depth, SGE limits and queue-pair type were unspecified, and which fields a transition read was unstated
uaf_modify_qp takes an attribute mask	

Changes from v2.1.3

Change	Reason
[R-9.5-002] states the window check as $\text{remote_addr} > \text{cxl_size}$ or $L > \text{cxl_size} - \text{remote_addr}$, and uaf_range_within() is the single spelling used everywhere	The inequality $\text{remote_addr} + L > \text{cxl_size}$ is the right mathematical rule and the wrong C: the sum wraps, so $\text{remote_addr} = 2^{64} - 8$ with $L = 16$ passes a naive test
[R-9.5-004] drops the [R-11.2-004] throttle and names the bounds check as the control	That counter counts packets that passed a link authenticator. The CXL path has neither packets nor a source address, so neither the per-QP nor the per-source counter can advance on it
The MAY alternative in [R-11.2-008] is withdrawn	Covering data packets with the fabric key had no byte layout, so it was a normative option nothing could implement. The required behaviour never depended on it
[R-9.5-006] no longer says the offset is "recorded at queue-pair creation"	[R-9.5-009] says both ends compute it and no field carries it; the formula is the address

Change	Reason
Section 6.7's command-specific fallback row gains its name cell	The mapping was sitting in the Name column
[R-1.2-001] says "this document" rather than "v2.1"; Appendix B's payload reads v2.1.4	Stale version strings

Changes from v2.1.2

Change	Reason
path occupies byte 59 of the connection record; reserved shrinks to bytes 60..63	[R-4.5-005] required the field to reach the peer and the record had no field for it. A strict implementation would have rejected every record the reference serialiser produces, because byte 59 was reserved-and-must-be-zero
[R-5.5-012] written The requirement index is a build gate	The conformance suite cited it and it did not exist A citation is now distinguished from a definition, and a cited-but-undefined requirement, a duplicate definition, or a requirement defined in a change-log section fails the build. This defect was invisible because the index recorded a citation as though it were a definition
Index entries carry the numbered section, e.g. 4.6 Memory registration	Three sections are titled "Memory registration"; fifteen requirements were attributed to a change-log table
[R-5.5-004] exempts the CXL path from the MTU floor	A CXL record carries no MTU, and the floor rejected it
[R-9.5-007] inherits [R-6.3-003]: initial expected phase 1, flip on wrap	Without the initial value a consumer starting at phase 0 accepts every zeroed entry as UAF_OK — the bug the phase tag exists to prevent
[R-9.5-009] gives the completion-ring offset a formula	The offset was "recorded at queue-pair creation" and appeared in no struct
Section 6.7 status codes corrected against NVMe Base 2.0d	PRP Offset Invalid is generic 0x13, not 0x0B; Attempted Write to Read Only Range is command-specific 0x82, not generic 0x1E; media 0x81 is Unrecovered Read Error and the protection-information tags are 0x82-0x84
[R-5.5-014] states the clock assumption behind the replay window	A 2 s window assumes the two clocks agree and never said so
Appendix B sets path; [R-9.5-002] uses remote_addr; [R-9.5-003] says base mask plus the CXL row; Section 1.6 and the async-event non-goal point at the right places	Stale references

Changes from v2.1.1

Change	Reason
enum uaf_path added; uaf_qp_init_attr.path and uaf_conn_info.path select the data plane	UAF-D declares Profile B for node-to-node traffic and also runs an intra-host CXL path with no MTU. Nothing selected between them, and [R-9.5-003] and [R-10.1-002] required different RTR masks for the same queue pair
Section 10.1 has three RTR columns: IBV, UDP, CXL	The CXL path has no MTU and no UDP address vector to require

Change	Reason
uaf_qp_init_attr grows to 40 bytes CM MAC input defined per body size; replay keyed on the header qp_id	The path field and its reserved word CM_RTU and CM_REJ carry no connection record, so neither the old MAC input nor the old replay key existed for half the CM packets CM_REP has no reason field
CM_REP carries UAF_CM_REASON_SIMUL_OPEN in imm_data The endpoint identity is built from source address, source UDP port and qp_num	Destination fields rebuild the asymmetry the tie-break exists to remove, and the connection record carries no address under Profile B
struct uaf_rmt_cqe (32 bytes) defined; ordering is a phase byte	[R-9.5-005] named no object: uaf_wc is host-side and has no phase byte, and the choice between a phase byte and an acquire on head was left open
rkey bound to a queue pair at connect by uaf_key_bind_qp(); an unbound rkey never validates	A queue pair does not exist at registration, so [R-11.2-003] described a state the samples never reached
Per-source-address throttling added ([R-11.2-008])	With no link authenticator, [R-11.2-004]'s counter never advances and a 32-bit key is scannable in seconds again
UAF_ACK_REQ_INTERVAL, UAF_ACK_COALESCE_NS, UAF_RKEY_THROTTLE_NS listed in uaf_config.h Version is 2.1.2 in the document, the header and CMake; SOVERSION is major.minor	All three were named by requirements and absent from the header The poll and create_qp signatures changed, so the three strings must agree and the shared object must carry a minor-level ABI number

Changes from v2.1

Change	Reason
uaf_storage_poll() fills struct uaf_storage_wc, not struct uaf_storage_cqe	[R-6.5-001] required the library to restore the caller's 64-bit wr_id, and the only identifier in the 16-byte ring image is a 16-bit cmd_id. The requirement had nowhere to land
CSPRNG and key-distinctness rules scoped to the software key table	[R-10.2-002] forbade UAF-N from synthesising keys while [R-10.2-001] required every backend to use the table and [R-11.2-002] required lkey != rkey, which verbs commonly violates. No UAF-N implementation could satisfy all three
UAF_OP_NAK split into NAK_SEQ (0x91), NAK_RNR (0x92), NAK_INVALID (0x93)	One opcode whose imm_data was already the expected PSN could not also express responder-not-ready
UAF_WIRE_FLAG_ECN_ECHO (bit 6) added; reserved mask now 0xFF80 ACK schedule added ([R-5.6-008], [R-5.6-009])	The receiver was told to reflect ECN CE in the next ACK, and an ACK had no field able to carry it A sender that never set ACK_REQ never advanced a Go-Back-N window
Simultaneous open decided by a canonical endpoint identity	Comparing (dest_ip, dest_udp_port, qp_num) made each peer compare a different tuple, so both could stay active or both go passive
CM packet length is per opcode: 96 for REQ and REP, 32 for RTU and REJ	[R-5.5-001] required 96 for every CM packet while the handshake diagram showed RTU carrying an authenticator only
enum uaf_cm_reason values assigned	UAF_CM_REASON_CLOSE was used and never given a value
An rkey failure limit now throttles; it never changes queue-pair state	Moving the queue pair to ERR after 16 failures handed an off-path host a connection kill in 16 datagrams, contradicting [R-5.7-001]

Change	Reason
UAF_QPT_UD enumerator withdrawn; UAF_CAP_MT_POST withdrawn	Both were named and never specified (OI-4, OI-5)
Atomic completion semantics moved to Section 4.11	They lived in Section 5, which [R-5.1-001] restricts to Profile B, so a portable caller could not rely on them for UAF-N

1 Introduction

1.1 Purpose

This document specifies the **Unified Accelerator Fabric (UAF)**, a portable C11 framework that provides two services across heterogeneous accelerator platforms:

- **Remote Memory Transport (RMT)** — direct memory access between compute nodes without CPU mediation in the data path.
- **Direct Storage Transport (DST)** — accelerator-initiated storage I/O to NVMe devices.

1.2 Scope

UAF specifies three platform backends.

Backend	Silicon	RMT data plane	DST data plane
UAF-N	NVIDIA H100/B200/GB200 with ConnectX-7/8	InfiniBand Verbs RC over IB or RoCEv2	cuFile (GPUDirect Storage)
UAF-S	Snapdragon X Elite, 8 Gen 3-4 (Linux)	UAF-RMT over UDP; PCIe P2P or FastRPC for local paths	NVMe via <code>io_uring</code> with <code>dma-heap</code> buffers
UAF-D	Dragonfly C1000	CXL.mem load/store with a software ring	NVMe over a CXL-mapped BAR

Non-goals

[R-1.2-001] The following are explicitly **out of scope** for this document. An implementation MUST NOT claim conformance on the basis of any of them.

- **Windows.** DirectStorage 1.4 is a Windows C++ API and was listed in v2.0's UAF-S row beside Linux `dma-heap` and FastRPC, which are not the same operating environment. A Windows binding SHALL be a separate document, UAF-SPEC-002. This document specifies Linux only.
- **Unreliable and multicast service.** Only the reliable connected queue pair is defined, and enum `uaf_qp_type` contains only `UAF_QPT_RC`. v2.1 declared a `UAF_QPT_UD` and never specified it; the enumerator is withdrawn (OI-4) rather than left as a hole for an implementer to guess at.
- **Shared receive queues, memory windows, on-demand paging, and re-registration.**
- **Interrupt-driven completion.** `UAF_CAP_CQ_EVENTS` is declared so a backend can advertise it, and the event API is deferred to v2.2. All required paths are polled.
- **An asynchronous event channel** for link-down and queue-pair fatal notification. Deferred to v2.2; see Appendix D, issue OI-1.

- **A fabric manager, subnet manager, or address resolution service.** Connection information reaches a peer by means outside this specification.
- **DOCA GPUNetIO.** v2.0's scope table named it and mapped it to nothing. It is a possible future path for a UAF-N implementation to speak Profile B; see Appendix D, issue OI-2.

Operating environment

[R-1.2-002] An implementation SHALL support Linux 5.15 or later on x86_64 or aarch64. UAF-D additionally requires a kernel with device-DAX (CONFIG_DEV_DAX) and CXL support (CONFIG_CXL_BUS). UAF-S requires CONFIG_DMABUF_HEAPS and a FastRPC-capable DSP stack.

[R-1.2-003] **One process, one backend.** A process SHALL load exactly one backend library. The core library resolves a single backend at `uaf_init()` time (Section 4.10); it MUST NOT link more than one, and the behaviour of a process that loads two is undefined.

1.3 Design principles

1. **Single C11 ABI.** An application compiles once against `uaf_core.h` and links against one backend (`libuaf_n.so`, `libuaf_s.so`, or `libuaf_d.so`).
2. **No CPU in the data path where the hardware permits.** UAF-N and UAF-D keep the host CPU out of the steady-state data path. UAF-S uses zero-copy SMMU-mapped rings or DSP offloads and does not make this claim for its UDP path.
3. **Zero-copy by default.** Every buffer path uses in-place registration of caller memory, or memory the library itself allocated. No path in this specification copies a payload in order to register it.
4. **Fail-explicit.** Every function reports a typed error. No silent degradation, no silent truncation, and no silent relocation.
5. **The portable surface is the API, not the packet.** This replaces v2.0's claim that one wire format spans all three backends, which could not be true while UAF-N was also offloaded to ConnectX hardware. An implementation declares a **wire profile** (Section 3.2); interoperability is defined *within* a profile, and Profile B over UDP is the common denominator available to every backend.

1.4 Normative language

[R-1.4-001] The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, MAY and OPTIONAL are to be interpreted as described in RFC 2119.

[R-1.4-002] Requirements are identified as [R-`<section>`-`<nnn>`] and are indexed in Appendix C. A conformance claim SHALL cite the requirement identifiers it satisfies.

[R-1.4-004] A paragraph that **begins** with a bracketed identifier is that requirement's single **definition**; every other occurrence of an identifier is a **citation**. Appendix C is generated from the definitions, and the build fails if any identifier is cited without being defined, is defined twice, or is defined outside a numbered section. v2.1.2 drew no such distinction, and a requirement that the conformance suite cited and no one had written — [R-5.5-012] — was indexed as though it existed and survived a whole revision.

[R-1.4-003] **Normative versus informative.** Tables of field offsets, requirement statements, and the headers reproduced in Appendix A are NORMATIVE. Every other code listing in Sections 7 through 9 is INFORMATIVE: it illustrates one way to satisfy the requirements and is not itself a requirement. v2.0 placed its backend listings in the body with no such marking, which made each of their defects arguably binding.

1.5 References

- RFC 2119, *Key words for use in RFCs to Indicate Requirement Levels*
- RFC 3720 Appendix B.4, *CRC32C (Castagnoli) definition and test vectors*
- InfiniBand Architecture Specification Vol. 1, Release 1.4
- Compute Express Link Specification, Revision 3.1 (PCIe 6.0 PHY, 64 GT/s)
- Compute Express Link Specification, Revision 4.0 (PCIe 7.0 PHY, 128 GT/s)
- NVM Express Base Specification, Revision 2.0d
- Microsoft DirectStorage 1.4 API Reference (out of scope; see Section 1.2)
- Qualcomm FastRPC User Guide (80-NH713-1)
- Linux rdma-core v50, libibverbs, <linux/dma-heap.h>, <linux/dma-buf.h>

1.6 Approval record

[R-1.6-001] This document SHALL NOT carry an “Approved for Implementation” status until every row below is complete. v2.0 carried that status with no approver, no review record and no open-issues list.

Role	Name	Date	Signature
Specification editor			
RMT protocol reviewer			
DST protocol reviewer			
UAF-N implementation owner			
UAF-S implementation owner			
UAF-D implementation owner			
Security reviewer			
Performance and conformance reviewer			

Review record. v2.0 received two independent engineering reviews, both dated 2026-09-29, and were followed by reviews of v2.1, v2.1.1 and v2.1.2. Every finding is dispositioned in Appendices E through I, newest first. Open items are carried in Appendix D.

2 Terminology and Naming

Term	Definition
RMT	Remote Memory Transport — RDMA-equivalent service
DST	Direct Storage Transport — accelerator-initiated NVMe I/O
UAF	Unified Accelerator Fabric
PAL	Platform Abstraction Layer
QP	Queue Pair (send and receive queues)
CQ	Completion Queue (RMT)
STQ	Storage Queue (DST submission). v2.0 called both the send queue and the storage queue “SQ”, in adjacent sections of one backend
CR	Completion Ring (DST)
SGE	Scatter-Gather Element
WR	Work Request
WC	Work Completion

+-----V-----+	+-----V-----+	+-----V-----+
ConnectX-7/8	PCIe / SMMU	CXL Root Port
H100/B200 GPU	Adreno / Hexagon	NVMe over CXL
+-----+	+-----+	+-----+

3.2 Wire profiles

[R-3.2-001] An implementation SHALL declare its wire profiles in `uaf_query_device()`'s `wire_profiles` field. Two profiles are defined.

Profile	Bit	Wire format	Reliability and congestion control
A (UAF_PROFILE_IBV)	1 << 0	IB transport headers (BTH, RETH) over InfiniBand or RoCEv2	Hardware: IB RC sequencing, ACK and retransmission; DCQCN offloaded to the adapter
B (UAF_PROFILE_UAFR)	1 << 1	The 64-byte UAFR header of Section 5 over UDP	Software: Sections 5.6 and 5.9

[R-3.2-002] A UAF-N implementation SHALL declare Profile A. It MUST NOT claim that a ConnectX adapter emits the UAFR header: the adapter emits IB transport headers, and the hardware reliable-connected engine supplies its own PSN, ACK and retransmission. It MAY additionally declare Profile B, and if it does, the Profile B path SHALL NOT claim `UAF_CAP_HW_CC`.

[R-3.2-003] A UAF-S implementation SHALL declare Profile B.

[R-3.2-004] A UAF-D implementation SHALL declare Profile B for node-to-node traffic. Its intra-host path is CXL.mem load/store through a software ring (Section 9.2) and carries no packet header at all; CXL.mem cannot carry a UAFR packet.

[R-3.2-005] Two implementations interoperate if and only if they share a profile. A conformance claim SHALL name the profile it was tested under.

Why this changed. v2.0 principle 5 asserted that UAF-RMT packets interoperate across all three backends over RoCEv2/UDP, while Section 5.6 simultaneously offloaded UAF-N congestion control to ConnectX hardware and Section 7.1 registered memory with `ibv_reg_mr`. Those cannot all hold: hardware offload means the IB transport is on the wire, and the `0x55414652` header is not. Neither can CXL.mem carry a UDP packet, nor is DMA-BUF a fabric. The portable surface is the C API.

3.3 CXL generation

[R-3.3-001] The UAF-D baseline SHALL be **CXL 3.1 on a PCIe 6.0 PHY at 64 GT/s**. An implementation targeting a PCIe 7.0 PHY at 128 GT/s SHALL cite **CXL 4.0** and declare it explicitly, because CXL 3.1 does not define that PHY. v2.0's scope table paired "CXL 3.1" with "PCIe Gen7", which is one generation apart, and its performance table then quoted a PCIe 7.0 line rate against a CXL 3.1 link.

[R-3.3-002] `UAF_ENABLE_CXL_CACHE` defaults to 0, and a build with it clear SHALL NOT advertise CXL.cache in any capability or document. v2.0's scope table listed CXL.cache as a UAF-D transport while the build flag that enables it defaulted to off. Where host-coherent atomics are required, CXL.cache is REQUIRED; see [R-5.4-006].

3.4 Compile-time configuration (uaf_config.h)

[R-3.4-001] A feature flag in uaf_config.h MUST NOT change the size or layout of any ABI struct. Every such struct is pinned by a `_Static_assert` (Section 4.1), so a build that violates this fails to compile.

[R-3.4-002] The resource limits in uaf_config.h are build-time defaults. An application MUST obtain the limits it relies on from `uaf_query_device()`. v2.0 baked `UAF_MAX_SGE`, `UAF_MAX_INLINE` and the queue depths into the application with no runtime query at all, so an application and a library built from different headers would disagree silently.

```
#define UAF_VERSION_MAJOR      2
#define UAF_VERSION_MINOR     1
#define UAF_VERSION_PATCH     4

/* Resource limits (defaults; query at runtime) */
#define UAF_MAX_QP             4096
#define UAF_MAX_CQ             4096
#define UAF_MAX_MR             65536
#define UAF_MAX_SGE           16
#define UAF_MAX_INLINE        256
#define UAF_CQ_DEPTH_DEFAULT  1024
#define UAF_SQ_DEPTH_DEFAULT  256
#define UAF_IS_POW2(x)         (((x) != 0u) && (((x) & ((x) - 1u)) == 0u))

/* Wire protocol (normative) */
#define UAF_WIRE_MAGIC         0x55414652U /* "UAFR" */
#define UAF_WIRE_VERSION      2
#define UAF_WIRE_HDR_SIZE     64

/* PATH MTU: counts the 64-byte header. v2.0 counted payload only, so a
 * maximum packet was 4160 bytes and did not fit a 4 KiB path MTU. */
#define UAF_WIRE_MTU_DEFAULT  4096
#define UAF_WIRE_MAX_SEG(mtu) ((uint32_t)((mtu) - UAF_WIRE_HDR_SIZE))
#define UAF_WIRE_SEG_DEFAULT  UAF_WIRE_MAX_SEG(UAF_WIRE_MTU_DEFAULT) /* 4032 */

/* Reliability (Section 5.6) */
#define UAF_RTO_BASE_NS        50000ULL /* 50 us */
#define UAF_RTO_MAX_NS        2000000ULL /* 2 ms */
#define UAF_RETRY_MAX          8
#define UAF_WINDOW_DEFAULT    256 /* segments */

/* ACK scheduling (Section 5.6). Named by [R-5.6-008] and [R-5.6-009]; v2.1.1
 * left both out of this header. */
#define UAF_ACK_REQ_INTERVAL   (UAF_WINDOW_DEFAULT / 2) /* 128 segments */
#define UAF_ACK_COALESCE_NS    25000ULL /* 25 us */

/* Security (Section 11) */
#define UAF_RKEY_FAIL_MAX      16
#define UAF_RKEY_THROTTLE_NS  1000000ULL /* 1 ms, [R-11.2-004] */
#define UAF_CM_NONCE_BYTES     8
#define UAF_CM_MAC_BYTES       16
#define UAF_CM_REPLAY_WINDOW_NS 2000000000ULL

#define UAF_CQE_ALIGN          16
```

4 Core Data Structures and ABI

4.1 ABI rules

[R-4.1-001] Every struct that crosses the library boundary SHALL use fixed-width integer types only. No enum field and no `size_t` field is permitted. In C11 an enum's size and signedness are implementation-defined and diverge under `-fshort-enums` and across compilers; `size_t` differs between ILP32 and LP64 Snapdragon userspace. v2.0 used both, in `uaf_wr.opcode`, `uaf_wc.status`, `uaf_qp_attr.qp_state` and `uaf_storage_cmd.buf_len`.

[R-4.1-002] Every such struct SHALL be pinned by a `_Static_assert` on its size in the public header, so layout drift is a compile error.

[R-4.1-003] A **configuration or query** struct the application fills, and that the library may later extend, SHALL carry `struct_size` as its first field, set by the caller: `uaf_mr_t`, `uaf_qp_init_attr`, `uaf_qp_attr`, `uaf_device_attr`, `uaf_conn_info`, `uaf_peer_info`, `uaf_storage_cmd`. The library SHALL fill only the prefix it understands and SHALL rewrite `struct_size` with what it filled. A **per-operation** struct on the hot path SHALL NOT carry it: `uaf_sge`, `uaf_wr`, `uaf_wc`, `uaf_storage_cqe`, `uaf_storage_wc`. Their layout is fixed for the life of an ABI major version and pinned by assertion, so a per-call size field would cost a branch on every work request and buy nothing. v2.1 applied the field unevenly and did not state which rule it followed.

[R-4.1-004] No struct in `uaf_types.h` is a wire format. Wire formats are byte arrays at fixed offsets, defined in Sections 5.2 and 5.5.

[R-4.1-005] A public struct MUST NOT expose backend-private state. v2.0's `uaf_mr.pal_priv` put a backend pointer in the application ABI and froze every backend's internals. A backend SHALL instead embed `uaf_mr_t` as the first member of its own private struct and recover it by cast.

The pinned sizes, all verified by conformance case C-1:

Struct	Size (bytes)	Alignment
<code>uaf_mr_t</code>	48	8
<code>struct uaf_sge</code>	16	8
<code>struct uaf_wr</code>	64	8
<code>struct uaf_wc</code>	40	8
<code>struct uaf_qp_init_attr</code>	40	4
<code>struct uaf_qp_attr</code>	80	8
<code>struct uaf_device_attr</code>	64	8
<code>struct uaf_conn_info</code>	64	8
<code>struct uaf_peer_info</code>	40	8
<code>struct uaf_storage_cmd</code>	56	8
<code>struct uaf_storage_cqe</code>	16	4
<code>struct uaf_storage_wc</code>	24	8
<code>struct uaf_rmt_cqe</code>	32	8

4.2 Handles and the memory region

```
typedef struct uaf_device uaf_device_t;
typedef struct uaf_domain uaf_domain_t;
typedef struct uaf_qp uaf_qp_t;
typedef struct uaf_cq uaf_cq_t;
typedef struct uaf_sq uaf_sq_t; /* storage queue (STQ) */
typedef struct uaf_cr uaf_cr_t; /* DST completion ring */
```

```

/* pal_priv is deliberately absent; see [R-4.1-005]. A backend does:
 * struct uaf_s_mr { uaf_mr_t pub; int dma_buf_fd; ... };
 * #define uaf_mr_priv(m) ((struct uaf_s_mr *) (m))
 */
typedef struct uaf_mr {
    uint32_t    struct_size;
    uint32_t    lkey;
    uint32_t    rkey;
    uint32_t    reserved0;
    uaf_domain_t *pd;
    void        *addr;
    uint64_t    length;
    uint64_t    flags;
} uaf_mr_t;
_Static_assert(sizeof(uaf_mr_t) == 48, "uaf_mr layout is ABI");

```

4.3 Enumerations

```

enum uaf_qp_state {
    UAF_QPS_RESET = 0, UAF_QPS_INIT = 1, UAF_QPS_RTR = 2, UAF_QPS_RTS = 3,
    UAF_QPS_SQD   = 4, UAF_QPS_SQE   = 5, UAF_QPS_ERR = 6,
};

enum uaf_wr_opcode {                /* LOCAL work request opcodes */
    UAF_WR_SEND           = 0x01,
    UAF_WR_SEND_INLINE    = 0x02,
    UAF_WR_RECV           = 0x03, /* local post only; NOT a wire opcode */
    UAF_WR_RDMA_READ      = 0x04,
    UAF_WR_RDMA_WRITE     = 0x05,
    UAF_WR_RDMA_WRITE_IMM = 0x06,
    UAF_WR_ATOMIC_CMP_SWP = 0x07,
    UAF_WR_ATOMIC_FETCH_ADD = 0x08,
    UAF_WR_STORAGE_READ   = 0x10,
    UAF_WR_STORAGE_WRITE  = 0x11,
    UAF_WR_STORAGE_FLUSH  = 0x12,
};

/* Wire-only opcodes. v2.0 used 0x80..0x91 on the wire and defined them in
 * no enum, so two implementations had nothing to agree on. */
enum uaf_wire_opcode {
    UAF_OP_CM_REQ  = 0x80, UAF_OP_CM_REP = 0x81,
    UAF_OP_CM_RTU  = 0x82, UAF_OP_CM_REJ = 0x83,
    UAF_OP_RDMA_READ_RESP = 0x84,
    UAF_OP_ATOMIC_ACK   = 0x88,
    UAF_OP_ACK          = 0x90,
    /* v2.1 had one NAK whose imm_data was already committed to the expected
     * PSN, while [R-5.6-006] also required a responder-not-ready NAK. */
    UAF_OP_NAK_SEQ = 0x91, /* out of sequence; expected PSN in imm_data */
    UAF_OP_NAK_RNR = 0x92, /* responder not ready; no receive buffer */
    UAF_OP_NAK_INVAL = 0x93, /* malformed request, e.g. misaligned atomic */
};

enum uaf_error {
    UAF_OK          = 0,
    UAF_ERR_INVAL   = -1, UAF_ERR_NOMEM          = -2,

```

```

    UAF_ERR_NODEV          = -3,  UAF_ERR_PERM          = -4,
    UAF_ERR_BUSY           = -5,  UAF_ERR_TIMEOUT       = -6,
    UAF_ERR_QP_STATE       = -7,
    UAF_ERR_RESERVED_8    = -8,  /* was UAF_ERR_CQ_EMPTY; retired */
    UAF_ERR_MR_FAULT       = -9,  UAF_ERR_NOT_SUPPORTED = -10,
    UAF_ERR_PROTO          = -11, UAF_ERR_CRC           = -12,
    UAF_ERR_REMOTE         = -13,
    UAF_ERR_RKEY           = -14, /* unknown, or not bound to this QP */
    UAF_ERR_AUTH           = -15, /* CM authentication or replay check */
    UAF_ERR_RELOCATED      = -16, /* cannot register in place */
};

enum uaf_wire_profile { UAF_PROFILE_IBV = 1u << 0, UAF_PROFILE_UAFR = 1u << 1 };

/* The data plane a queue pair uses. A profile is not enough to say which:
 * UAF-D declares UAF_PROFILE_UAFR for node-to-node traffic and also runs an
 * intra-host CXL.mem path with no packet header and no MTU. */
enum uaf_path {
    UAF_PATH_IBV = 1, /* Profile A: IB or RoCEv2 transport */
    UAF_PATH_UDP = 2, /* Profile B: UAF-RMT header over UDP */
    UAF_PATH_CXL = 3, /* CXL.mem ring, intra-host (Section 9.5) */
};

enum uaf_device_cap {
    UAF_CAP_ATOMICS      = 1ull << 0, UAF_CAP_DST          = 1ull << 1,
    UAF_CAP_GPU_MR       = 1ull << 2, UAF_CAP_CXL_MR       = 1ull << 3,
    UAF_CAP_DMABUF_MR    = 1ull << 4, UAF_CAP_CQ_EVENTS    = 1ull << 5,
    UAF_CAP_REG_ANY_MEM  = 1ull << 6, UAF_CAP_SYNC_FAULT    = 1ull << 7,
    UAF_CAP_HW_CC        = 1ull << 8,
    /* Bit 9 reserved. v2.1 referenced a UAF_CAP_MT_POST that appeared in no
     * enum and had no defined ring discipline (OI-5); it is withdrawn. */
    UAF_CAP_RESERVED_9   = 1ull << 9,
};

enum uaf_send_flags {
    UAF_SEND_SIGNALED    = 1u << 0, /* raise a work completion */
    UAF_SEND_INLINE      = 1u << 1, /* payload is inline, <= max_inline_data */
    UAF_SEND_SOLICITED   = 1u << 2, /* solicited event at the responder */
    UAF_SEND_FENCE       = 1u << 3, /* do not start before prior WRs complete */
};

enum uaf_mr_flags {
    UAF_MR_LOCAL_WRITE   = 1ull << 0, UAF_MR_REMOTE_WRITE = 1ull << 1,
    UAF_MR_REMOTE_READ   = 1ull << 2, UAF_MR_ATOMIC       = 1ull << 3,
    UAF_MR_GPU           = 1ull << 4, UAF_MR_CXL           = 1ull << 5,
};

/* v2.1 used UAF_CM_REASON_CLOSE and assigned it no value. */
enum uaf_cm_reason {
    UAF_CM_REASON_CLOSE   = 0x00, UAF_CM_REASON_NO_QP      = 0x01,
    UAF_CM_REASON_MTU_MISMATCH = 0x02, UAF_CM_REASON_PROFILE = 0x03,
    UAF_CM_REASON_AUTH      = 0x04, UAF_CM_REASON_RESOURCE  = 0x05,
    UAF_CM_REASON_STALE_CONN = 0x06, UAF_CM_REASON_SIMUL_OPEN = 0x07,
};

```

4.3.1 Connection and peer structures

Both were pinned in the size table of Section 4.1 and their layouts were absent from v2.1.

```
/* Host-local. NOT the exchange format: it has interior padding and host byte  
 * order. Serialise with uaf_conn_info_serialize() ([R-5.5-002]). */  
struct uaf_conn_info {  
    uint32_t struct_size;  
    uint32_t qp_num;  
    uint32_t psn;  
    uint32_t rkey;  
    uint64_t remote_va;  
    uint64_t cxl_base;  
    uint64_t cxl_size;  
    uint8_t gid[16];  
    uint16_t lid;  
    uint16_t mtu; /* bytes, including the header */  
    uint8_t sl;  
    uint8_t traffic_class;  
    uint8_t wire_profile; /* exactly one enum uaf_wire_profile bit */  
    uint8_t path; /* enum uaf_path; MUST agree with the peer */  
}; /* 64 bytes */  
  
/* Same-host peer mapping. dma_buf_fd MUST arrive out of band via SCM_RIGHTS  
 * over an AF_UNIX socket; it is meaningless to a remote node ([R-5.5-005]). */  
struct uaf_peer_info {  
    uint32_t struct_size;  
    uint32_t rkey;  
    uint64_t remote_va;  
    uint64_t length;  
    uint64_t cxl_base;  
    int32_t dma_buf_fd; /* -1 if unused */  
    uint32_t reserved0;  
}; /* 40 bytes */
```

4.4 Work requests and completions

```
struct uaf_sge { uint64_t addr; uint32_t length; uint32_t lkey; };  
  
struct uaf_wr {  
    uint64_t wr_id;  
    uint32_t opcode; /* enum uaf_wr_opcode */  
    uint32_t num_sge;  
    struct uaf_sge *sge;  
    uint32_t send_flags; /* enum uaf_send_flags */  
    uint32_t imm_data;  
    uint64_t remote_addr;  
    uint32_t rkey;  
    uint32_t reserved0;  
    uint64_t compare_add;  
    uint64_t swap;  
};  
  
struct uaf_wc {  
    uint64_t wr_id;  
    uint32_t opcode;  
    int32_t status; /* enum uaf_error */  
};
```

```

uint32_t byte_len;
uint32_t qp_num;
uint32_t src_qp;
uint32_t imm_data;
uint32_t wc_flags;          /* UAF_WC_* */
uint32_t reserved0;
};

#define UAF_WC_WITH_IMM      (1U << 0) /* imm_data is valid */
#define UAF_WC_SOLICITED    (1U << 1)
#define UAF_WC_ATOMIC_ORIG  (1U << 2) /* SGE 0 holds the pre-operation value */

```

[R-4.4-001] `imm_data` in a work completion is valid if and only if `UAF_WC_WITH_IMM` is set. v2.0 delivered `imm_data` with no way to know whether it was present.

[R-4.4-002] **Ownership at post time.** `uaf_post_send()` and `uaf_post_recv()` SHALL copy the work request *and* the `num_sge` elements of the SGE array before returning. The caller MAY reuse or free both immediately. The buffers the SGEs point to remain owned by the caller until the corresponding completion is reaped. v2.0 never stated this, and its UAF-S sample copied the work request while leaving `sge` pointing at caller memory.

[R-4.4-003] `num_sge` SHALL be at most the `max_send_sge` or `max_recv_sge` agreed at `uaf_create_qp()`, which SHALL be at most `uaf_query_device()`'s `max_sge`. A larger value SHALL fail with `UAF_ERR_INVALID`.

[R-4.4-004] **Gather is local.** Under Profile B the wire carries one `remote_va` and one contiguous byte stream; a sender walks its SGE list and emits that stream. The wire carries no SGE list. Under Profile A the adapter performs gather natively. An application observes the same semantics either way.

4.5 Queue pair creation and modification

```

/* Reliable connected is the only type v2.1.1 defines. v2.1 declared an
 * unreliable datagram type and never specified it (OI-4). */
enum uaf_qp_type { UAF_QPT_RC = 1 };

struct uaf_qp_init_attr {
    uint32_t struct_size;
    uint32_t qp_type;
    uint32_t max_send_wr;          /* MUST be a power of two */
    uint32_t max_recv_wr;          /* MUST be a power of two */
    uint32_t max_send_sge;
    uint32_t max_recv_sge;
    uint32_t max_inline_data;
    uint32_t sq_sig_all;          /* 0: only UAF_SEND_SIGNALED WRs complete */
    uint32_t path;                /* enum uaf_path */
    uint32_t reserved0;
};

enum uaf_qp_attr_mask {
    UAF_QP_STATE      = 1u<<0,   UAF_QP_SQ_PSN      = 1u<<1,
    UAF_QP_RQ_PSN     = 1u<<2,   UAF_QP_DEST_QPN     = 1u<<3,
    UAF_QP_PATH_MTU   = 1u<<4,   UAF_QP_AV_GID       = 1u<<5,
    UAF_QP_AV_LID     = 1u<<6,   UAF_QP_AV_UDP       = 1u<<7,
    UAF_QP_AV_CXL     = 1u<<8,   UAF_QP_MAX_RD_ATOMIC = 1u<<9,
    UAF_QP_MAX_DEST_RD_AT = 1u<<10, UAF_QP_RETRY      = 1u<<11,
    UAF_QP_RNR_RETRY  = 1u<<12,  UAF_QP_PORT         = 1u<<13,
}

```

```

    UAF_QP_TIMEOUT      = 1u<<14,
};

struct uaf_qp_attr {
    uint32_t struct_size;
    uint32_t qp_state;
    uint32_t sq_psn;
    uint32_t rq_psn;
    uint32_t dest_qp_num;
    uint32_t path_mtu;          /* BYTES, including the 64-byte header */
    uint16_t dest_lid;
    uint8_t  max_rd_atomic;
    uint8_t  max_dest_rd_atomic;
    uint8_t  retry_cnt;
    uint8_t  rnr_retry;
    uint8_t  port_num;
    uint8_t  reserved0;
    uint8_t  dest_gid[16];      /* Profile A */
    uint8_t  dest_ip[16];      /* Profile B, IPv4-mapped permitted */
    uint16_t dest_udp_port;
    uint16_t src_udp_port;
    uint32_t rnr_timer_ns;
    uint64_t timeout_ns;       /* 0 selects UAF_RTO_BASE_NS */
};

```

[R-4.5-005] `uaf_qp_init_attr.path` SHALL name the data plane the queue pair uses, and `uaf_conn_info.path` SHALL carry the same value to the peer. A connection whose two ends disagree SHALL be rejected with `UAF_ERR_PROTO`. The required RTR attribute mask depends on this field, not on the profile (Section 10.1). v2.1.1 had no such field: UAF-D declares Profile B and also runs an intra-host CXL path, so `uaf_create_qp()` and `uaf_conn_info` between them could not say which data plane a queue pair was for.

[R-4.5-001] `uaf_create_qp()` SHALL take a struct `uaf_qp_init_attr`. v2.0's `create_qp` took two completion queues and nothing else: no depth, no SGE limit, no queue-pair type, so two implementations would allocate differently from the same call.

[R-4.5-002] `path_mtu` is expressed in **bytes and includes the 64-byte header**. v2.0 declared it `uint16_t` and its sample assigned 4096, while a verbs backend needs the IB enumeration (`IBV_MTU_4096 = 5`); the unit was never stated. A backend SHALL convert to its native encoding internally and SHALL reject a value that is not a supported path MTU.

[R-4.5-003] `uaf_modify_qp()` SHALL take an attribute mask. A transition SHALL fail with `UAF_ERR_INVALID` if a field it requires is absent from the mask. The required masks are in Section 10.1.

[R-4.5-004] `attr->qp_state` SHALL equal the state argument. A mismatch SHALL fail with `UAF_ERR_INVALID` rather than silently preferring one. v2.0's reference application set both and the specification never said which won.

4.6 Memory registration

[R-4.6-001] `uaf_reg_mr()` SHALL register the caller's memory **in place**. On success (`*mr`)->`addr == addr`. An implementation MUST NOT copy, bounce or relocate the buffer.

[R-4.6-002] A backend that cannot map arbitrary caller memory SHALL fail `uaf_reg_mr()` with `UAF_ERR_NOT_SUPPORTED` and SHALL clear `UAF_CAP_REG_ANY_MEM`.

[R-4.6-003] `uaf_alloc_mr()` SHALL allocate DMA-capable memory and register it in one step, returning the chosen address in `(*mr) ->addr`. Every backend SHALL implement it. A portable application either tests `UAF_CAP_REG_ANY_MEM` or always allocates through `uaf_alloc_mr()`.

[R-4.6-004] `uaf_dereg_mr()` releases a region from `uaf_reg_mr()`; `uaf_free_mr()` releases one from `uaf_alloc_mr()`. Crossing them SHALL fail with `UAF_ERR_INVALID`.

Why this changed. v2.0's `sd_reg_mr()` allocated a second buffer, memcpy'd the caller's bytes into it, and reported the copy's address. Two consequences, neither documented: every later write the application made to its own buffer was invisible to the fabric, and "zero-copy by default" was false on that backend. The reference application in v2.0 Appendix B did exactly that and would have transmitted stale data.

4.7 Return conventions

[R-4.7-001] A **control** function returns `UAF_OK` (0) on success or a negative enum `uaf_error` on failure.

[R-4.7-002] A **poll** function — `uaf_poll_cq()` and `uaf_storage_poll()` — returns a non-negative count of entries written, from 0 to max inclusive, or a negative enum `uaf_error`. A return of 0 means the queue was empty and is not an error.

[R-4.7-003] A caller SHALL test `n < 0` for failure. Testing `n != UAF_OK` is incorrect and would treat a successful batch of one as an error.

[R-4.7-004] `UAF_ERR_CQ_EMPTY` is retired and its value (−8) reserved. No function returns it.

Why this changed. v2.0 stated as principle 4 that every API returns a typed error, while `poll_cq(cq, max, wc)` must also report how many completions it produced — and `UAF_OK` is 0, which is also "filled none". Its reference application looped on `== UAF_ERR_CQ_EMPTY`, a reading that forecloses the batch polling max exists for. The single most-used function on the data path had no defined contract.

4.8 Device capability query

```
struct uaf_device_attr {
    uint32_t struct_size;      /* caller sets to sizeof(...) */
    uint32_t wire_profiles;    /* enum uaf_wire_profile bitmask */
    uint64_t caps;             /* enum uaf_device_cap bitmask */
    uint32_t max_qp;           uint32_t max_cq;
    uint32_t max_mr;           uint32_t max_sge;
    uint32_t max_inline_data;  uint32_t max_cqe;
    uint32_t max_sqe;          uint32_t path_mtu_max;
    uint64_t max_msg_len;
    uint8_t max_rd_atomic;     uint8_t max_dest_rd_atomic;
    uint8_t reserved0[6];
};

int uaf_query_device(uaf_device_t *dev, struct uaf_device_attr *attr);
```

[R-4.8-001] `uaf_query_device()` is the only authoritative source of a device's limits and capabilities. v2.0 had no such call; an application could not discover max SGE, max inline, path MTU, atomic support, or even whether DST existed on the backend it was linked against.

[R-4.8-002] An application SHALL NOT use an OPTIONAL service whose capability bit is clear. Doing so SHALL fail with `UAF_ERR_NOT_SUPPORTED`.

4.9 Threading

[R-4.9-001] A queue pair, completion queue, storage queue and completion ring are **single-producer, single-consumer**. Concurrent `uaf_post_send()` on one queue pair from two threads, or concurrent `uaf_poll_cq()` on one completion queue, is undefined. There is no capability that makes it defined: v2.1.1's `UAF_CAP_MT_POST` is withdrawn and bit 9 of the capability word is reserved (OI-5). An application needing concurrency SHALL use one queue pair per thread, which [R-4.9-002] permits without external locking.

[R-4.9-002] Distinct queue pairs MAY be used concurrently from distinct threads without external locking.

[R-4.9-003] `uaf_init()`, `uaf_fini()`, `uaf_query_device()` and the registration calls are thread-safe.

v2.0 had no threading section. Both of its ring samples were plainly single-producer — load the producer index, compute the next, store it, with no compare-and-swap — and nothing said so, so a correct-looking multi-threaded application would have silently corrupted a ring.

4.10 Backend loading

[R-4.10-001] Each backend library SHALL export exactly one symbol, `const struct uaf_pal_ops *uaf_pal_get_ops(void)`.

[R-4.10-002] The public headers MUST NOT declare per-backend symbols. v2.0's `uaf_pal.h` declared `uaf_pal_nvidia`, `uaf_pal_snapdragon` and `uaf_pal_dragonfly` together, so any consumer of the header had to link all three backends or take undefined symbols.

[R-4.10-003] `uaf_init()` SHALL resolve exactly one backend: the value of the `UAF_BACKEND` environment variable if set, otherwise the single backend library present. If none or more than one candidate is found, `uaf_init()` SHALL fail with `UAF_ERR_NODEV`.

4.11 Observable completion semantics

These rules are **profile-independent** and bind every backend. They live here rather than in Section 5 because [R-5.1-001] restricts Section 5 to Profile B, and an application must be able to rely on them on UAF-N too. v2.1 placed the atomic result rule in Section 5.4, which left a portable caller with no stated behaviour for Profile A.

[R-4.11-001] **Atomic result delivery.** On completion of `UAF_WR_ATOMIC_CMP_SWP` or `UAF_WR_ATOMIC_FETCH_ADD`, the value the target held **before** the operation SHALL be present in the first 8 bytes of SGE 0 of the originating work request, and the completion SHALL carry `UAF_WC_ATOMIC_ORIG` with `byte_len = 8`. SGE 0 SHALL be at least 8 bytes and locally writable; a work request that does not satisfy this SHALL fail with `UAF_ERR_INVALID`. The wire encoding that realises this under Profile B is in Section 5.4; a Profile A implementation obtains the same observable result through the IB transport.

[R-4.11-002] **Completion count.** Exactly one work completion SHALL be raised per work request that requested one, regardless of how many packets, segments or descriptors the implementation used.

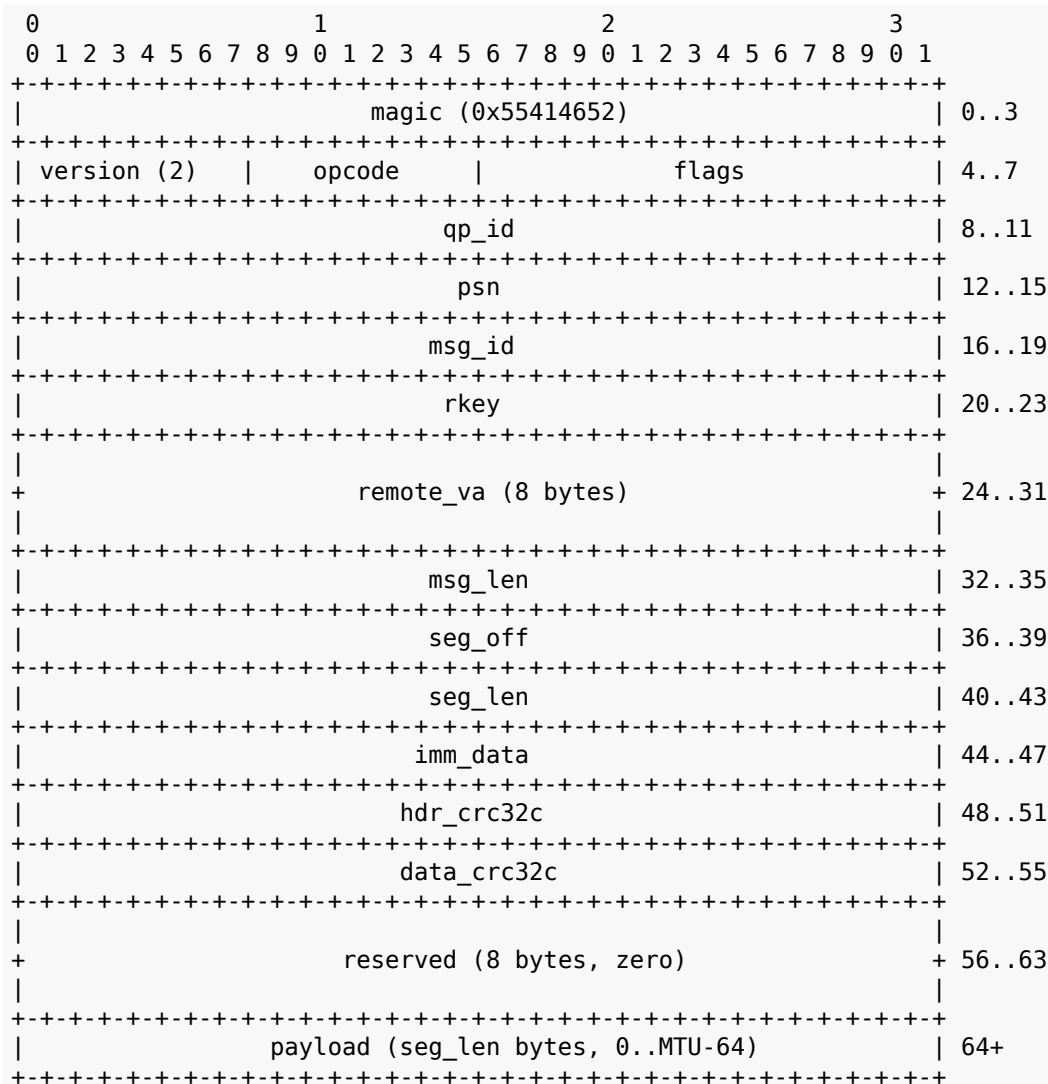
[R-4.11-003] **byte_len.** For a SEND or an RDMA WRITE, `byte_len` is the number of payload bytes transferred. For an RDMA READ it is the number of bytes placed locally. For an atomic it is 8.

5 RMT Protocol — Profile B (UAF-RMT over UDP)

[R-5.1-001] This section applies to UAF_PROFILE_UAFR only. A Profile A implementation follows the InfiniBand Architecture Specification for its wire behaviour and uses this document only for the API of Section 4.

5.1 Packet format

[R-5.1-002] All multi-byte wire fields SHALL be encoded in network byte order. The fixed header SHALL be 64 bytes.



5.2 Field definitions

Field	Offset	Size	Description
magic	0	4	0x55414652 ("UAFR")
version	4	1	Protocol version, 2

Field	Offset	Size	Description
opcode	5	1	enum uaf_wr_opcode or enum uaf_wire_opcode, per Section 5.3
flags	6	2	See below; bits 7..15 reserved and MUST be zero
qp_id	8	4	Destination queue-pair number
psn	12	4	Packet sequence number, incremented per packet
msg_id	16	4	Identifies the message a segment belongs to
rkey	20	4	32-bit remote key
remote_va	24	8	Base target address of the message , 8-byte aligned in the header
msg_len	32	4	Total length of the message in bytes
seg_off	36	4	Byte offset of this segment within the message
seg_len	40	4	Payload bytes carried by this packet
imm_data	44	4	Meaning depends on the opcode: immediate data on a WRITE_IMM or SEND; the cumulative PSN acknowledged on an ACK; the expected PSN on a NAK_SEQ; a retry delay in microseconds on a NAK_RNR; an enum uaf_error on a NAK_INVALID; an enum uaf_cm_reason on a CM_REJ or a simultaneous-open CM_REP
hdr_crc32c	48	4	CRC32C over header bytes 0..47
data_crc32c	52	4	CRC32C over the seg_len payload bytes; valid when UAF_WIRE_FLAG_DATA_CRC is set
reserved	56	8	MUST be zero on transmit and MUST be checked on receive

Flag	Bit	Meaning
UAF_WIRE_FLAG_INLINE	0	Payload is inline data, at most UAF_MAX_INLINE bytes
UAF_WIRE_FLAG_IMM	1	imm_data is meaningful
UAF_WIRE_FLAG_SOLICITED	2	Solicited event at the responder
UAF_WIRE_FLAG_ACK_REQ	3	Responder SHALL send an ACK

Flag	Bit	Meaning
UAF_WIRE_FLAG_DATA_CRC	4	data_crc32c is valid
UAF_WIRE_FLAG_RETRY	5	This packet is a retransmission
UAF_WIRE_FLAG_ECN_ECHO	6	On an ACK: CE-marked data arrived in the interval this ACK covers

[R-5.2-001] remote_va is the base address of the whole message. The target address of a segment is remote_va + seg_off. This keeps every segment self-describing without a second address field.

[R-5.2-002] A receiver SHALL reject a header whose reserved bytes or reserved flag bits are non-zero, with UAF_ERR_PROT0. The reserved area lies outside hdr_crc32c's coverage, so this check is explicit and not implied by the CRC.

v2.0's header spent 4 bytes on a reserved lkey beside 20 further reserved bytes, carried a single length with no message identity and no segment offset, and left the payload outside every integrity check.

5.3 Opcode semantics

[R-5.3-001] The following, and only the following, are wire opcodes.

Opcode	Value	Requires rkey	Payload
UAF_WR_SEND	0x01	No	0..MTU-64 bytes of the message
UAF_WR_SEND_INLINE	0x02	No	1..256 bytes; single segment only
UAF_WR_RDMA_READ	0x04	Yes	Request: none. msg_len is the length requested
UAF_WR_RDMA_WRITE	0x05	Yes	seg_len bytes written at remote_va + seg_off
UAF_WR_RDMA_WRITE_IMM	0x06	Yes	As WRITE; the last segment also delivers imm_data
UAF_WR_ATOMIC_CMP_SWP	0x07	Yes	16 bytes: compare, then swap
UAF_WR_ATOMIC_FETCH_ADD	0x08	Yes	8 bytes: addend
UAF_OP_RDMA_READ_RESP	0x84	No	seg_len bytes of the requested data
UAF_OP_ATOMIC_ACK	0x88	No	8 bytes: the pre-operation value
UAF_OP_CM_REQ / REP / RTU / REJ	0x80..0x83	No	Section 5.5
UAF_OP_ACK	0x90	No	None; imm_data carries the cumulative PSN acknowledged
UAF_OP_NAK_SEQ	0x91	No	None; imm_data carries the expected PSN
UAF_OP_NAK_RNR	0x92	No	None; imm_data carries the retry delay in μ s

Opcode	Value	Requires rkey	Payload
UAF_OP_NAK_INVALID	0x93	No	None; imm_data carries an enum uaf_error

[R-5.3-002] UAF_WR_RECV (0x03) SHALL NOT appear on the wire. It is a local receive-buffer post. v2.0's opcode table listed it as a packet with an empty payload, which gave implementers a packet to build that has no meaning. Conformance case C-3 rejects it at both encode and decode.

5.3.1 Framing

[R-5.3-003] A message of msg_len bytes SHALL be cut into segments of at most MTU - 64 bytes. Every segment of one message SHALL carry the same msg_id, msg_len and remote_va.

[R-5.3-004] FIRST and LAST SHALL be **derived**, never signalled:

$$\text{FIRST} \equiv (\text{seg_off} = 0), \quad \text{LAST} \equiv (\text{seg_off} + \text{seg_len} = \text{msg_len})$$

A sender therefore cannot contradict itself, and there is no flag for a receiver to disbelieve.

[R-5.3-005] A receiver SHALL raise **exactly one** work completion per message, on the LAST segment, with byte_len = msg_len. If the bytes accepted do not total msg_len, the message SHALL be completed with UAF_ERR_PROTO.

[R-5.3-006] A zero-length message occupies exactly one segment with msg_len = seg_len = 0.

[R-5.3-007] seg_off + seg_len SHALL NOT exceed msg_len. A violation is UAF_ERR_PROTO on receive and UAF_ERR_INVALID on send.

v2.0 could not express a message larger than one packet. Every packet carried remote_va and length with no message identity, no total length, no segment offset and no first/middle/last indication, so a 1 MiB RDMA WRITE had no encoding and a responder had no way to know when to raise one completion.

5.4 Atomics

[R-5.4-001] Atomic operands SHALL be 8 bytes, big-endian.

Packet	Opcode	seg_len	Payload
Compare-and-swap request	0x07	16	compare (8) then swap (8)
Fetch-and-add request	0x08	8	addend (8)
Atomic acknowledgement	0x88	8	value before the operation (8)

[R-5.4-002] remote_va for an atomic SHALL be 8-byte aligned. A responder receiving a mis-aligned atomic SHALL NOT perform the operation and SHALL reply UAF_OP_NAK_INVALID (0x93) with UAF_ERR_INVALID in imm_data; the local API rejects it with UAF_ERR_INVALID.

[R-5.4-003] The acknowledgement SHALL echo the request's msg_id.

[R-5.4-004] The requester SHALL write the returned value into SGE 0 of the originating work request, which SHALL be at least 8 bytes and locally writable, and SHALL set UAF_WC_ATOMIC_ORIG with byte_len = 8. A work request whose SGE 0 is smaller than 8 bytes SHALL fail with UAF_ERR_INVALID.

[R-5.4-005] Atomics are atomic with respect to other UAF atomics on the same responder, and with respect to RDMA READ and WRITE targeting the same 8 bytes.

[R-5.4-006] Atomicity with respect to **host CPU** accesses on UAF-D REQUIRES CXL.cache and therefore UAF_ENABLE_CXL_CACHE. A build without it SHALL clear UAF_CAP_ATOMICS.

[R-5.4-007] max_dest_rd_atomic bounds the outstanding atomic and READ requests a responder will accept. v2.0 had max_rd_atomic and no responder-side limit at all.

v2.0 specified a 16-byte request payload and never defined the response. The fetched value had nowhere to go: uaf_wc had no field for it and no local SGE semantics were stated. Atomics were unimplementable.

5.5 Connection management

[R-5.5-001] A CM packet SHALL consist of the 64-byte header of Section 5.1 with a CM opcode, followed by the body below. msg_len SHALL equal seg_len, and seg_len SHALL be exactly:

Opcode	Body	seg_len
UAF_OP_CM_REQ	64-byte connection record + 32-byte authenticator	96
UAF_OP_CM_REP	64-byte connection record + 32-byte authenticator	96
UAF_OP_CM_RTU	32-byte authenticator only	32
UAF_OP_CM_REJ	32-byte authenticator only; reason in imm_data	32

A CM packet whose seg_len disagrees with this table SHALL be dropped with UAF_ERR_PROTO. v2.1 required 96 for every CM packet while its own handshake diagram showed RTU carrying an authenticator only.

5.5.1 Connection record (64 bytes, big-endian)

Field	Offset	Size	Description
qp_num	0	4	Sender's queue-pair number
psn	4	4	Sender's initial PSN
rkey	8	4	Remote key for the region being offered
lid	12	2	Profile A only; zero under Profile B
mtu	14	2	Path MTU in bytes, including the header
remote_va	16	8	Base of the offered region
cxl_base	24	8	UAF-D CXL window offset
cxl_size	32	8	UAF-D CXL window length
gid	40	16	Profile A GID; zero under Profile B
sl	56	1	Service level
traffic_class	57	1	Traffic class
wire_profile	58	1	Exactly one enum uaf_wire_profile bit
path	59	1	enum uaf_path; the data plane both ends will use
reserved	60	4	MUST be zero

[R-5.5-002] struct `uaf_conn_info` is a host-local struct with interior padding and host byte order. It SHALL NOT be written to a socket. An implementation SHALL use `uaf_conn_info_serialize()` and `uaf_conn_info_deserialize()`. v2.0's `uaf_conn_info` had 6 bytes of interior padding, no byte-order rule, and was evidently meant to be memcpy'd to a peer.

[R-5.5-003] `wire_profile` SHALL name exactly one profile. Zero, or more than one bit, is `UAF_ERR_PROTO`.

[R-5.5-004] On a packet-carrying path — `UAF_PATH_IBV` or `UAF_PATH_UDP` — `mtu` SHALL exceed 64, because a smaller value leaves no room for the header; a violation is `UAF_ERR_PROTO`. The floor SHALL NOT be applied to `UAF_PATH_CXL`, which carries no packet and no MTU.

[R-5.5-012] The record's `path` field SHALL carry the enum `uaf_path` value of [R-4.5-005], and a receiver SHALL reject a record whose `path` is not one of the three defined values, or is inconsistent with `wire_profile`, with `UAF_ERR_PROTO`:

path	Permitted <code>wire_profile</code>
<code>UAF_PATH_IBV</code>	<code>UAF_PROFILE_IBV</code>
<code>UAF_PATH_UDP</code>	<code>UAF_PROFILE_UAFR</code>
<code>UAF_PATH_CXL</code>	<code>UAF_PROFILE_UAFR</code>

v2.1.2 added `path` to `uaf_conn_info` and to `uaf_qp_init_attr`, required it to be carried to the peer, and left the record's byte 59 reserved-and-zero. The field never reached the wire, so the discriminant did not survive the only byte exchange this document defines, and a strict implementation would have rejected every record the reference serialiser produced. Conformance case C-9 cites this requirement, which is how the gap was found — after the citation had been in the document for a revision with nothing behind it.

[R-5.5-014] The replay window of [R-5.5-007] assumes the two endpoints' clocks agree to within `UAF_CM_REPLAY_WINDOW_NS`. A deployment SHALL keep them synchronised to a tighter bound than that window — NTP is sufficient for the 2 s default, PTP for a shorter one — and an implementation SHOULD count and log timestamp rejections, because a clock that has drifted out of the window presents as a total connection failure with `UAF_ERR_AUTH`.

[R-5.5-005] A connection record SHALL NOT contain a file descriptor. A descriptor is a process-local integer: copied to another node it is meaningless, and interpreted locally it is a confused-deputy bug. v2.0 carried `dma_buf_fd` in `uaf_conn_info`. Descriptors now appear only in struct `uaf_peer_info`, which is same-host only and whose descriptor SHALL be transferred out of band with `SCM_RIGHTS` over an `AF_UNIX` socket.

5.5.2 Authenticator (32 bytes)

Field	Offset	Size	Description
<code>nonce</code>	0	8	Sender's random nonce, from a CSPRNG
<code>timestamp_ns</code>	8	8	Sender's clock, nanoseconds
<code>mac</code>	16	16	HMAC-SHA256, truncated to 128 bits

[R-5.5-006] The MAC SHALL be computed, keyed by the pre-shared fabric key of [R-11.2-007], over the concatenation below, which **depends on the body size**:

A side builds its own identity from the address and port it sends from, and the peer's identity from the source address and port of the IP/UDP header of the CM_REQ it received, together with the qp_num in that packet. Destination fields SHALL NOT be used: A's destination port is B's source port, so using them rebuilds the asymmetry this rule exists to remove. v2.1.1 fixed the comparison and still did not say which fields fill the image — and the connection record zeroes the GID under Profile B and carries no IP address, so the answer was not available inside it. If both sides send CM_REQ before either receives one, each side compares the two identities by memcmp of that image; the side whose own identity compares **lower** SHALL continue as the active side, and the other SHALL abandon its own CM_REQ and answer with CM_REP carrying UAF_CM_REASON_SIMUL_OPEN in imm_data; CM_REP has no reason field of its own, and CM_REJ uses imm_data the same way. Because both peers know both identities and evaluate the same comparison, the outcome is complementary by construction. Equal identities cannot arise and SHALL be reported as UAF_ERR_PROTO.

v2.1 compared (dest_ip, dest_udp_port, qp_num). Each side's *destination* is the other side, so the two peers compared different tuples: both could conclude they were lower and stay active, or both could conclude they were higher and go passive. Conformance case C-11 asserts that $\text{is_active}(a,b) + \text{is_active}(b,a) == 1$ for every distinguishing field.

[R-5.5-010] **Teardown.** uaf_disconnect() SHALL send CM_REJ with reason UAF_CM_REASON_CLOSE (0x00, see enum uaf_cm_reason in Section 4.3) in imm_data, flush outstanding work with UAF_ERR_QP_STATE, and move the queue pair to RESET. A CM_REJ received on an established connection SHALL be treated identically.

[R-5.5-011] Retransmission of CM_REQ, CM_REP and CM_RTU SHALL use the timers of Section 5.6. An unanswered CM_REQ after UAF_RETRY_MAX attempts SHALL fail uaf_connect() with UAF_ERR_TIMEOUT.

v2.0's Section 5.4 was a three-arrow diagram. There was no packet layout, no address format, no authentication, no simultaneous-open rule, no teardown, no reject, and the opcodes it named appeared in no enum. uaf_connect() existed in the header and the reference application never called it.

5.6 Reliability

[R-5.6-001] Profile B SHALL use Go-Back-N with cumulative acknowledgement. psn increments once per packet and wraps modulo 2^{32} .

[R-5.6-002] A sender that may be bridged to Profile A SHALL keep psn within 24 bits, because the IB transport carries a 24-bit PSN.

[R-5.6-003] A receiver SHALL drop a packet whose psn is not the expected one and SHOULD reply UAF_OP_NAK_SEQ (0x91) with the expected PSN in imm_data.

[R-5.6-004] The send window SHALL be at least the bandwidth-delay product of the path, in segments:

$$W_{\min} = \left\lceil \frac{R \cdot \text{RTT}}{8 \cdot L_{\text{seg}}} \right\rceil$$

for line rate R bit/s, round trip RTT seconds and segment payload L_{seg} bytes. The default UAF_WINDOW_DEFAULT is 256 segments, which covers 800 Gbps at a 10 μs round trip with 4032-byte segments ($W_{\min} = 248$).

[R-5.6-005] The retransmit timeout SHALL start at UAF_RT0_BASE_NS (50 μs), double on each expiry, and saturate at UAF_RT0_MAX_NS (2 ms). After UAF_RETRY_MAX expires the queue

pair SHALL move to UAF_QPS_ERR and the outstanding work SHALL complete with UAF_ERR_TIMEOUT.

[R-5.6-006] rnr_retry and rnr_timer_ns govern the responder-not-ready case: a responder with no posted receive buffer SHALL reply UAF_OP_NAK_RNR with a retry delay in microseconds in imm_data, and the requester SHALL wait the larger of that delay and rnr_timer_ns and retry up to rnr_retry times before failing with UAF_ERR_TIMEOUT. v2.0 declared rnr_retry and gave it no behaviour; v2.1 gave it behaviour and no opcode to carry it.

[R-5.6-008] **ACK schedule, sender side.** A sender SHALL set UAF_WIRE_FLAG_ACK_REQ on the LAST segment of every message and at least once every UAF_ACK_REQ_INTERVAL segments, which defaults to half the window, 128 segments. Without this a sender that never sets the flag never learns what has been received and a Go-Back-N window never advances — v2.1 stated no schedule at all.

[R-5.6-009] **ACK schedule, receiver side.** A receiver SHALL emit an ACK on any segment carrying ACK_REQ, and otherwise within UAF_ACK_COALESCE_NS (25 μ s by default) of receiving in-order data. An ACK SHALL carry the highest contiguous PSN received in imm_data.

[R-5.6-010] A receiver that observed the IP ECN CE codepoint on any data packet covered by an ACK SHALL set UAF_WIRE_FLAG_ECN_ECHO on that ACK. This is the signal F_k of Section 5.9 is measured from; v2.1 told the receiver to reflect CE “in the next ACK” and gave the ACK nothing to reflect it with.

[R-5.6-007] A Profile A implementation SHALL NOT run these timers. IB RC retransmission is performed by the adapter, and a second software timer over it would produce duplicate retransmission. v2.0 specified one software timer for all backends while also claiming hardware offload on UAF-N.

v2.0 specified a 100 ms base timeout in a fabric whose stated target is 1.5 μ s — roughly 6×10^4 round trips — and a fixed 64-segment window, which caps a 400 Gbps link at about 205 Gbps once the round trip reaches 10 μ s.

5.7 Error handling

Condition	Action	Reported as
magic mismatch	Drop silently; queue-pair state unchanged	—
version unknown	Drop silently; state unchanged	—
hdr_crc32c mismatch	Drop; count; state unchanged	UAF_ERR_CRC
data_crc32c mismatch	Drop; do not write the payload; NAK	UAF_ERR_CRC
Reserved bytes or flags non-zero	Drop	UAF_ERR_PROTO
Unexpected psn	Drop; NAK_SEQ with the expected PSN	—
Unknown or unbound rkey	Drop; count against UAF_RKEY_FAIL_MAX	UAF_ERR_RKEY
Access outside the region	Drop; complete the request	UAF_ERR_MR_FAULT
Access flags insufficient	Drop; complete the request	UAF_ERR_PERM
No receive buffer posted	NAK_RNR with a retry delay; do not consume the segment	UAF_ERR_TIMEOUT after rnr_retry attempts
CM MAC or replay check failed	Drop	UAF_ERR_AUTH
CM body length wrong for the opcode	Drop	UAF_ERR_PROTO
Atomic target misaligned	NAK_INVALID; do not perform the operation	UAF_ERR_INVALID

[R-5.7-001] A malformed or unauthenticated packet MUST NOT move a queue pair to UAF_QPS_ERR. Otherwise any host able to reach the UDP port could terminate a connection with one datagram.

5.8 Test vectors

[R-5.8-001] An implementation SHALL reproduce the following values. They are produced and checked by conformance case C-2.

CRC32C is Castagnoli, reflected, polynomial 0x1EDC6F41 (reflected 0x82F63B78), initial value 0xFFFFFFFF, final XOR 0xFFFFFFFF.

Input	CRC32C
32 bytes of 0x00	0x8A9136AA
32 bytes of 0xFF	0x62A8AB43
32 bytes, 0x00 through 0x1F	0x46DD794E
"123456789"	0xE3069283
"a"	0xC1D04330
empty	0x00000000

Golden header. An RDMA WRITE, opcode 0x05, flags = ACK_REQ | DATA_CRC (0x0018), qp_id 0x00000101, psn 1, msg_id 7, rkey 0xDEADBEEF, remote_va 0x0000100000000000, msg_len 16, seg_off 0, seg_len 16, imm_data 0, payload the 16 ASCII bytes UAF-SPEC-001 v21:

```
00: 55 41 46 52 02 05 00 18 00 00 01 01 00 00 00 01
16: 00 00 00 07 DE AD BE EF 00 00 10 00 00 00 00 00
32: 00 00 00 10 00 00 00 00 00 00 00 10 00 00 00 00
48: CD 6B 9B 1A 89 50 EC 50 00 00 00 00 00 00 00 00
```

hdr_crc32c = 0xCD6B9B1A over bytes 0..47. data_crc32c = 0x8950EC50 over the 16 payload bytes.

Connection record. qp_num 0x01020304, psn 0x00ABCDEF, rkey 0xCAFEBAFE, lid 0x1234, mtu 4096, remote_va 0x00001000000002000, cxl_base 0x0000200000000000, cxl_size 0x0000000040000000, gid 0xE0..0xEF, sl 3, traffic_class 7, wire_profile 2 (UAF_PROFILE_UAFR) and path 2 (UAF_PATH_UDP) serialise to:

```
00: 01 02 03 04 00 AB CD EF CA FE BA BE 12 34 10 00
16: 00 00 10 00 00 00 20 00 00 00 20 00 00 00 00 00
32: 00 00 00 00 40 00 00 00 E0 E1 E2 E3 E4 E5 E6 E7
48: E8 E9 EA EB EC ED EE EF 03 07 02 02 00 00 00 00
```

Byte 58 is the profile, byte 59 the path, and bytes 60..63 are the four remaining reserved bytes. Conformance case C-9 emits these bytes, checks the round trip, and checks that a host-struct memcpy does *not* produce them.

5.9 Congestion control

[R-5.9-001] A Profile A implementation SHALL use the adapter's DCQCN and SHALL declare UAF_CAP_HW_CC.

[R-5.9-002] A Profile B implementation SHALL implement the following rate controller. v2.0 gave two equations with α never defined or updated, no time constant, no R_{AT} , no minimum rate, and no statement of whether a mark is an IP ECN codepoint or a RoCE CNP — which is not enough for two implementations to interoperate without one starving the other.

Congestion is signalled by the IP ECN codepoint CE on a received data packet, which the receiver reflects with UAF_WIRE_FLAG_ECN_ECHO on the next ACK ([R-5.6-010]). F_k is the fraction of bytes acknowledged in interval k by ACKs carrying that flag. Let F_k be the fraction of acknowledged bytes marked in interval k , g the smoothing weight, T the update period, R_{AI} the additive-increase step and $R_{\min}$ the floor:

$$\alpha_k = (1 - g) \alpha_{k-1} + g F_k$$

$$R_{k+1} = \max\left(R_{\min}, R_k \left(1 - \frac{\alpha_k}{2}\right)\right) \quad \text{if } F_k > 0$$

$$R_{k+1} = \min(R_{\text{line}}, R_k + R_{AI}) \quad \text{if } F_k = 0$$

[R-5.9-003] Defaults SHALL be $g = 1/16$, $T = 50 \mu s$, $R_{AI} = R_{\text{line}}/256$, $R_{\min} = R_{\text{line}}/1024$.

[R-5.9-004] The rate SHALL be reduced at most once per update period T . Without this bound a burst of marks collapses the rate geometrically toward the floor.

6 DST Protocol

6.1 Structure of this section

v2.0 Section 6.1 specified a 64-byte “UAF SQE” with its own crc32c field. No backend filled it; every sample built an NVMe submission entry instead, and the coverage of that CRC was never stated. The pseudo-format is **deleted**. DST is specified here as a mapping from struct `uaf_storage_cmd` onto the NVMe I/O command set, which is what implementations actually emit.

[R-6.1-001] A DST implementation SHALL submit NVMe I/O commands as defined by NVM Express Base 2.0d. It SHALL NOT define an intermediate wire format.

[R-6.1-002] Queue creation, the admin queue, Identify Controller, Identify Namespace, namespace formatting and completion-queue association are preconditions and are outside this specification. `uaf_storage_open()` SHALL fail with `UAF_ERR_NODEV` if they are not satisfied. v2.0 assumed all of them silently.

6.2 Command mapping

```
struct uaf_storage_cmd {
    uint32_t struct_size;
    uint32_t opcode;           /* enum uaf_wr_opcode, STORAGE_* only */
    uint64_t wr_id;
    uint64_t lba;
    uint32_t nlb;             /* 1-BASED block count; 0 is invalid */
    uint32_t buf_offset;
    void *buf;
    uint64_t buf_len;         /* was size_t in v2.0: not ABI-stable */
    uint32_t flags;
    uint32_t reserved0;
};
```

[R-6.2-001] The opcode map SHALL be total. Every value not in the table is UAF_ERR_INVALID.

uaf_storage_cmd.opcode	NVMe opcode
UAF_WR_STORAGE_READ (0x10)	0x02 Read
UAF_WR_STORAGE_WRITE (0x11)	0x01 Write
UAF_WR_STORAGE_FLUSH (0x12)	0x00 Flush

v2.0's sample selected the opcode with a two-way ternary on READ, so UAF_WR_STORAGE_FLUSH fell through the WRITE arm and a flush was issued as a write. Conformance case C-8 asserts the flush mapping explicitly and asserts that it is not the write opcode.

[R-6.2-002] nlb is a **1-based** block count in the API. NVMe CDW12 carries a **0-based** count. The library performs the conversion, SHALL reject nlb == 0 with UAF_ERR_INVALID, and SHALL reject nlb > 65536.

nlb (API)	CDW12 NLB	Result
0	—	UAF_ERR_INVALID
1	0x0000	One block
8	0x0007	Eight blocks
65536	0xFFFF	Largest legal transfer
65537 or more	—	UAF_ERR_INVALID

v2.0 computed (cmd->nlb - 1) & 0xFFFF with no zero check, so a zero-length request became a 65536-block transfer — 256 MiB at a 4 KiB block size. Neither the API nor the diagram said whether the on-ring value was 0-based or 1-based.

[R-6.2-003] The data pointer SHALL be selected from the buffer's IOVA and length against the controller's memory page size. A single PRP entry describes at most one page-boundary crossing.

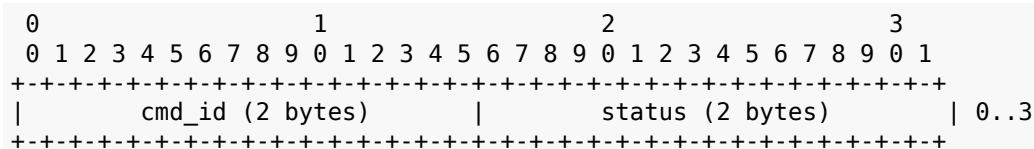
Condition	Data pointer
Buffer ends within the first page	PRP1 only
Buffer crosses exactly one page boundary	PRP1 and PRP2
More than two pages, controller without SGL support	PRP2 points at a PRP list
More than two pages, controller with SGL support	NVMe SGL

v2.0 assigned the whole buffer to dptr_prp1 regardless of length, which is correct only for a buffer that fits in one page.

[R-6.2-004] buf_len SHALL be at least nlb multiplied by the namespace block size, after buf_offset. A shorter buffer is UAF_ERR_INVALID.

6.3 Completion ring

[R-6.3-001] A completion ring entry SHALL be exactly 16 bytes, naturally aligned, and SHALL NOT be packed.





Field	Offset	Size	Description
cmd_id	0	2	Command identifier allocated by the library
status	2	2	int16_t, an enum uaf_error value
bytes_transferred	4	4	Bytes actually moved
latency_ns	8	4	Clamped at UINT32_MAX, never wrapped
reserved0	12	2	MUST be zero
reserved1	14	1	MUST be zero
phase	15	1	Bit 0 is the phase tag; bits 7..1 reserved

[R-6.3-002] phase SHALL be the **last byte** of the entry and SHALL be the only byte a consumer reads before the remainder is known valid. A producer SHALL write bytes 0..14, issue a release barrier, and then write phase.

[R-6.3-003] A ring SHALL be zero-filled at creation and the initial expected phase SHALL be 1. A consumer SHALL read phase, compare it with the expected phase, and only then read the rest of the entry. The expected phase SHALL flip on every wrap.

[R-6.3-004] A ring base SHALL be UAF_CQE_ALIGN (16-byte) aligned, and the depth SHALL be a power of two.

[R-6.3-005] The consumer SHALL read phase with **acquire** semantics. Reading it and then reading the body in program order is not sufficient: on AArch64 the body loads may be satisfied ahead of the phase load. v2.1 specified the producer's release store and gave the consumer no matching acquire, on the one path the phase tag exists to make safe.

6.3.1 The host completion

[R-6.3-006] The 16-byte entry above is the **device ring image**. What uaf_storage_poll() delivers is the host completion below, whose wr_id is the caller's full 64-bit value.

```

struct uaf_storage_wc {
    uint64_t wr_id;           /* the caller's value, restored in full */
    int32_t status;          /* enum uaf_error */
    uint32_t bytes_transferred;
    uint32_t latency_ns;
    uint32_t reserved0;
};                               /* 24 bytes */

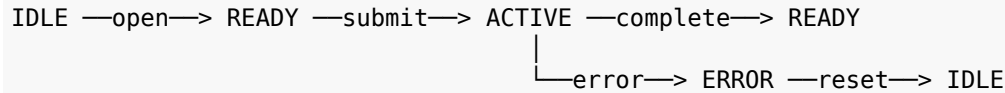
```

v2.1 required the library to restore the caller's wr_id ([R-6.5-001]) while giving uaf_storage_poll() only the 16-byte image, whose sole identifier is a 16-bit cmd_id. The internal table alone could not satisfy the public API: there was no field to write the answer into. Conformance case C-7 now submits 0x00000000000001234 and 0xAAAABBBB000001234 and requires both to arrive intact through uaf_storage_poll().

Why this matters. v2.0's CQE had 4 reserved bytes and no phase tag, and its status field encoded success as UAF_OK = 0 — the same bits a zeroed slot holds. A poller therefore could not distinguish a fresh completion from an untouched one, and DST completion polling was unimplementable. v2.0 also marked the struct packed,

which drops its alignment to 1, defeats the aligned 16-byte load a ring wants, and is undefined behaviour on strict-alignment targets. Conformance case C-7 polls a zeroed ring and requires 0 completions.

6.4 Storage queue state machine



[R-6.4-001] `uaf_storage_reset(sq, cr)` drives the ERROR to IDLE edge. v2.0 drew that edge with no entry point able to traverse it.

[R-6.4-002] `uaf_storage_close(sq, cr)` SHALL release **both** objects that `uaf_storage_open()` created. v2.0's close took only the submission queue, so the completion ring — and on UAF-N the `cuFile` batch handle it shares — was never released.

6.5 Command identifiers

[R-6.5-001] `cmd_id` SHALL be allocated by the library as an index into an in-flight table, and the library SHALL restore the caller's full 64-bit `wr_id` into struct `uaf_storage_wc` before delivering a completion.

[R-6.5-002] A `cmd_id` SHALL NOT be reused while its command is in flight. When the table is full, `uaf_storage_submit()` SHALL return `UAF_ERR_BUSY`.

[R-6.5-003] Releasing an unknown or already-free `cmd_id` is `UAF_ERR_PROTO`. A completion naming such a `cmd_id` SHALL cause `uaf_storage_poll()` to return `UAF_ERR_PROTO` rather than deliver a completion with an invented `wr_id`.

v2.0 defined `cmd_id` as the low 16 bits of `wr_id`. Two in-flight commands whose `wr_ids` differed only above bit 16 produced the same `cmd_id`, and since the completion carried nothing else, the application could not recover its own identifier. RMT delivered a full 64-bit `wr_id` in `uaf_wc`; DST silently narrowed it. Conformance case C-8 submits `0x00000000000001234` and `0xAAAABBBB000001234` together and requires both to come back intact.

6.6 Submission ring memory type

[R-6.6-001] A submission ring a device reads SHALL be either device-coherent, or explicitly cleaned to memory before the doorbell is written. A cacheable ring plus a bare doorbell is a stale-descriptor bug.

[R-6.6-002] Doorbell ordering follows Section 9.4.

6.7 NVMe completion status

[R-6.7-001] An NVMe completion status SHALL be mapped to an enum `uaf_error` and reported in `uaf_storage_wc.status`. v2.1 specified the opcode and NLB mapping and left completion status unmapped, so an implementation could submit a command and had no defined way to say why it failed.

SCT is the Status Code Type and SC the Status Code of the NVMe completion.

SCT	SC	Name in NVMe Base 2.0d	enum uaf_error
0 Generic	0x00	Successful Completion	UAF_OK
0 Generic	0x01	Invalid Command Opcode	UAF_ERR_INVAL
0 Generic	0x02	Invalid Field in Command	UAF_ERR_INVAL
0 Generic	0x03	Command ID Conflict	UAF_ERR_PROTO
0 Generic	0x04	Data Transfer Error	UAF_ERR_REMOTE
0 Generic	0x06	Internal Error	UAF_ERR_REMOTE
0 Generic	0x07	Command Abort Requested	UAF_ERR_BUSY
0 Generic	0x0B	Invalid Namespace or Format	UAF_ERR_INVAL
0 Generic	0x0D	Invalid SGL Segment Descriptor	UAF_ERR_MR_FAULT
0 Generic	0x0E	Invalid Number of SGL Descriptors	UAF_ERR_MR_FAULT
0 Generic	0x0F	Data SGL Length Invalid	UAF_ERR_MR_FAULT
0 Generic	0x13	PRP Offset Invalid	UAF_ERR_MR_FAULT
0 Generic	0x15	Operation Denied	UAF_ERR_PERM
0 Generic	0x16	SGL Offset Invalid	UAF_ERR_MR_FAULT
0 Generic	0x1E	SGL Data Block Granularity Invalid	UAF_ERR_MR_FAULT
0 Generic	0x20	Namespace Is Write Protected	UAF_ERR_PERM
0 Generic	0x80	LBA Out of Range	UAF_ERR_MR_FAULT
0 Generic	0x81	Capacity Exceeded	UAF_ERR_NOMEM
0 Generic	0x82	Namespace Not Ready	UAF_ERR_NODEV
0 Generic	0x83	Reservation Conflict	UAF_ERR_BUSY
1 Command-specific	0x81	Invalid Protection Information	UAF_ERR_CRC
1 Command-specific	0x82	Attempted Write to Read Only Range	UAF_ERR_PERM
1 Command-specific	any other	—	UAF_ERR_INVAL
2 Media	0x80	Write Fault	UAF_ERR_REMOTE
2 Media	0x81	Unrecovered Read Error	UAF_ERR_REMOTE
2 Media	0x82	End-to-end Guard Check Error	UAF_ERR_CRC
2 Media	0x83	End-to-end Application Tag Check Error	UAF_ERR_CRC
2 Media	0x84	End-to-end Reference Tag Check Error	UAF_ERR_CRC
2 Media	0x85	Compare Failure	UAF_ERR_REMOTE
2 Media	0x86	Access Denied	UAF_ERR_PERM
2 Media	0x87	Deallocated or Unwritten Logical Block	UAF_ERR_MR_FAULT
3 Path related	any	—	UAF_ERR_TIMEOUT
any	any unlisted	—	UAF_ERR_REMOTE

v2.1.2's table used three wrong code points against the specification it cites: 0x0B is Invalid Namespace or Format and PRP Offset Invalid is 0x13; Attempted Write to Read Only Range is command-specific 0x82, while generic 0x1E is SGL Data Block Granularity Invalid; and media 0x81 is Unrecovered Read Error, with the protection-information tags at 0x82-0x84. A faithful implementation of that table would have agreed with the document and disagreed with the controller. Conformance case C-8 now pins each code point, not merely that an unlisted one is negative.

[R-6.7-002] An unlisted status SHALL map to a failure. It SHALL NOT map to UAF_OK. Conformance case C-8 sweeps every (SCT, SC) pair other than generic-success and requires a negative result.

7 Platform Backend: UAF-N (NVIDIA)

Listings in this section are INFORMATIVE ([R-1.4-003]).

7.1 Profile and capabilities

[R-7.1-001] UAF-N SHALL declare UAF_PROFILE_IBV, UAF_CAP_HW_CC, and UAF_CAP_REG_ANY_MEM. It SHALL declare UAF_CAP_GPU_MR only if it implements the dma-buf registration path of [R-7.3-001].

7.2 Device discovery

[R-7.2-001] An ibv_device pointer obtained from ibv_get_device_list() SHALL NOT be used after ibv_free_device_list(). An implementation SHALL either open the device before freeing the list, or retain the list for the lifetime of its device array.

```
struct uaf_n_devs {
    struct ibv_device **list; /* retained: freeing it invalidates entries */
    int count;
};

static int nv_get_device_list(uaf_device_t ***devs, int *count)
{
    struct uaf_n_devs *s = calloc(1, sizeof(*s));
    if (!s) return UAF_ERR_NOMEM;

    s->list = ibv_get_device_list(&s->count);
    if (!s->list) { free(s); return UAF_ERR_NODEV; }
    if (s->count <= 0) { /* a non-NULL empty list still owns */
        ibv_free_device_list(s->list); /* memory; v2.1's sample leaked it */
        free(s);
        return UAF_ERR_NODEV;
    }

    uaf_device_t **out = calloc((size_t)s->count, sizeof(*out));
    if (!out) { ibv_free_device_list(s->list); free(s); return UAF_ERR_NOMEM; }

    for (int i = 0; i < s->count; i++) {
        struct uaf_n_dev *d = calloc(1, sizeof(*d));
        if (!d) { nv_unwind(out, i, s); return UAF_ERR_NOMEM; }
        d->ibv_dev = s->list[i]; /* valid while s->list is retained */
        d->owner = s;
        d->name = strdup(ibv_get_device_name(s->list[i]));
        if (!d->name) { free(d); nv_unwind(out, i, s); return UAF_ERR_NOMEM; }
        out[i] = &d->pub;
    }
    *devs = out; *count = s->count;
    return UAF_OK; /* s->list freed in free_device_list */
}
```

v2.0 stored `list[i]` into its device array and then called `ibv_free_device_list(list)`, leaving every retained pointer dangling. It also left `strdup` and each per-device `calloc` unchecked.

7.3 Memory registration

[R-7.3-001] Registering GPU memory SHALL use `ibv_reg_dmabuf_mr()` with a descriptor exported from CUDA, or the peer-memory client where the platform provides one. Plain `ibv_reg_mr()` on a CUDA device pointer is not the GPUDirect RDMA path.

[R-7.3-002] `UAF_MR_GPU` SHALL select that path. v2.0 defined the flag and used it nowhere, so it had no behaviour at all; `UAF_MR_CXL` was likewise unused.

[R-7.3-003] A registration failure SHALL be mapped from `errno`, not reported uniformly as `UAF_ERR_NOMEM`:

errno	UAF error
ENOMEM	UAF_ERR_NOMEM
EINVAL	UAF_ERR_INVALID
EACCES, EPERM	UAF_ERR_PERM
EFAULT	UAF_ERR_MR_FAULT
EOPNOTSUPP, ENOTSUP	UAF_ERR_NOT_SUPPORTED
anything else	UAF_ERR_NODEV

```
static int nv_reg_mr(uaf_domain_t *pd, void *addr, size_t len,
                    uint64_t flags, uaf_mr_t **out)
{
    int access = 0;
    if (flags & UAF_MR_LOCAL_WRITE) access |= IBV_ACCESS_LOCAL_WRITE;
    if (flags & UAF_MR_REMOTE_WRITE) access |= IBV_ACCESS_REMOTE_WRITE;
    if (flags & UAF_MR_REMOTE_READ) access |= IBV_ACCESS_REMOTE_READ;
    if (flags & UAF_MR_ATOMIC) access |= IBV_ACCESS_REMOTE_ATOMIC;
    access |= IBV_ACCESS_RELAXED_ORDERING; /* required for GPU peer DMA */

    struct uaf_n_mr *m = calloc(1, sizeof(*m));
    if (!m) return UAF_ERR_NOMEM;

    struct ibv_mr *ib;
    if (flags & UAF_MR_GPU) {
        int dmabuf_fd = -1;
        int rc = nv_export_cuda_dmabuf(addr, len, &dmabuf_fd); /* [R-7.3-001] */
        if (rc != UAF_OK) { free(m); return rc; }
        ib = ibv_reg_dmabuf_mr(pd->ibv_pd, 0, len,
                               (uint64_t)(uintptr_t)addr, dmabuf_fd, access);
        if (ib) m->dmabuf_fd = dmabuf_fd; else close(dmabuf_fd);
    } else {
        ib = ibv_reg_mr(pd->ibv_pd, addr, len, access);
    }
    if (!ib) { int rc = nv_errno_to_uaf(errno); free(m); return rc; }

    m->ibv_mr = ib; /* private; not in the public MR */
    m->pub.struct_size = (uint32_t)sizeof(uaf_mr_t);
    m->pub.pd = pd;
    m->pub.addr = addr; /* in place: [R-4.6-001] */
    m->pub.length = len;
    m->pub.lkey = ib->lkey;
}
```

```

m->pub.rkey      = ib->rkey;
m->pub.flags     = flags;
*out = &m->pub;
return UAF_OK;
}

```

7.4 DST via cuFile

[R-7.4-001] A UAF-N DST implementation SHALL call `cuFileDriverOpen()` once, SHALL register every data buffer with `cuFileBufRegister()`, SHALL submit with `cuFileBatchIOSubmit()`, and SHALL reap with `cuFileBatchIOGetStatus()`.

[R-7.4-002] The doorbell mechanism is internal to cuFile and the NVMe driver. This specification SHALL NOT describe it as a UAF-visible operation. v2.0's doorbell table claimed "cuFileBatchIOSubmit batches NVMe SQ tail updates via `nvidia-fs/BAR0`", which describes a kernel path rather than the API being specified.

```

static int nv_storage_open(uaf_domain_t *pd, const char *path,
                          uaf_sq_t **out_sq, uaf_cr_t **out_cr)
{
    if (nv_driver_open_once() != UAF_OK) return UAF_ERR_NODEV;

    int fd = open(path, O_RDWR | O_DIRECT | O_CLOEXEC);
    if (fd < 0) return nv_errno_to_uaf(errno);

    CUfileDescr_t desc = {0};
    desc.type          = CU_FILE_HANDLE_TYPE_OPAQUE_FD;
    desc.handle.fd     = fd;

    CUfileHandle_t fh;
    if (cuFileHandleRegister(&fh, &desc).err != CU_FILE_SUCCESS) {
        close(fd); return UAF_ERR_NODEV;
    }
    CUfileBatchHandle_t batch;
    if (cuFileBatchIOSetUp(&batch, UAF_SQ_DEPTH_DEFAULT).err
        != CU_FILE_SUCCESS) {
        cuFileHandleDeregister(fh); close(fd); return UAF_ERR_NOMEM;
    }

    struct uaf_n_sq *sq = calloc(1, sizeof(*sq));
    struct uaf_n_cr *cr = calloc(1, sizeof(*cr));
    if (!sq || !cr) {
        free(sq); free(cr); cuFileBatchIODestroy(batch);
        cuFileHandleDeregister(fh); close(fd); return UAF_ERR_NOMEM;
    }
    if (uaf_cid_init(&sq->cids, UAF_SQ_DEPTH_DEFAULT) != UAF_OK) {
        free(sq); free(cr); cuFileBatchIODestroy(batch);
        cuFileHandleDeregister(fh); close(fd); return UAF_ERR_NOMEM;
    }
    sq->fh = fh; sq->batch = batch; sq->fd = fd;
    cr->batch = batch; cr->owner = sq; /* closed together: [R-6.4-002] */
    *out_sq = &sq->pub; *out_cr = &cr->pub;
    return UAF_OK;
}

```

8 Platform Backend: UAF-S (Snapdragon, Linux)

Listings in this section are INFORMATIVE ([R-1.4-003]).

[R-8.1-001] UAF-S SHALL declare UAF_PROFILE_UAFR. It SHALL NOT declare UAF_CAP_HW_CC: its congestion control is the software controller of Section 5.9.

[R-8.1-002] This backend is Linux-only. DirectStorage is a Windows C++ API and is out of scope ([R-1.2-001]); v2.0 listed it in the same backend row as Linux dma-heap and FastRPC.

8.1 Memory registration

[R-8.2-001] sd_reg_mr() SHALL map the caller's pages for device access in place and SHALL NOT allocate a second buffer. If the SMMU mapping of caller memory is unavailable, it SHALL return UAF_ERR_NOT_SUPPORTED and the backend SHALL clear UAF_CAP_REG_ANY_MEM.

[R-8.2-002] sd_alloc_mr() SHALL allocate from /dev/dma_heap/system, map it, and register it. This is the portable path.

[R-8.2-003] A non-coherent buffer SHALL be synchronised with DMA_BUF_IOCTL_SYNC around every CPU access window. A barrier does not flush a dirty cache line.

[R-8.2-004] A length passed to the FastRPC layer SHALL be range-checked before narrowing. v2.0 passed (int)len from a size_t.

[R-8.2-005] Keys SHALL come from the CSPRNG key table of Section 11.2. A descriptor number SHALL NOT be used as a key, and a key SHALL NOT be derived from one.

```
struct uaf_s_mr { uaf_mr_t pub; int dma_buf_fd; int owns_mmap; };

static int sd_alloc_mr(uaf_domain_t *pd, size_t len, uint64_t flags,
                      uaf_mr_t **out)
{
    if (len == 0u || len > (size_t)INT_MAX) return UAF_ERR_INVALID;

    int heap_fd = open("/dev/dma_heap/system", O_RDWR | O_CLOEXEC);
    if (heap_fd < 0) return UAF_ERR_NODEV;

    struct dma_heap_allocation_data alloc = {
        .len = len, .fd_flags = O_RDWR | O_CLOEXEC,
    };
    if (ioctl(heap_fd, DMA_HEAP_IOCTL_ALLOC, &alloc) < 0) {
        close(heap_fd); return UAF_ERR_NOMEM;
    }
    close(heap_fd);

    void *va = mmap(NULL, len, PROT_READ | PROT_WRITE, MAP_SHARED,
                    alloc.fd, 0);
    if (va == MAP_FAILED) { close(alloc.fd); return UAF_ERR_NOMEM; }

    /* Map into the Hexagon SMMU context. Return value is checked. */
    int rc = remote_register_buf_attr(va, (int)len, alloc.fd,
                                      FASTRPC_ATTR_NON_COHERENT);
    if (rc != 0) { munmap(va, len); close(alloc.fd); return UAF_ERR_PERM; }

    struct uaf_s_mr *m = calloc(1, sizeof(*m));
    if (!m) { munmap(va, len); close(alloc.fd); return UAF_ERR_NOMEM; }

    uint32_t lkey = 0, rkey = 0;
```

```

/* Registered unbound: uaf_connect() installs the queue-pair component of
 * the binding, because no queue pair exists yet ([R-11.2-003]). */
rc = uaf_key_register(&pd->keys, (uint64_t)(uintptr_t)va, len, flags,
                    0u, &lkey, &rkey); /* [R-8.2-005] */
if (rc != UAF_OK) {
    free(m); munmap(va, len); close(alloc.fd); return rc;
}

m->dma_buf_fd = alloc.fd;
m->owns_mmap = 1;
m->pub.struct_size = (uint32_t)sizeof(uaf_mr_t);
m->pub.pd = pd;
m->pub.addr = va; /* the allocator chose it: [R-4.6-003] */
m->pub.length = len;
m->pub.lkey = lkey;
m->pub.rkey = rkey;
m->pub.flags = flags;
*out = &m->pub;
return UAF_OK;
}

```

v2.0's `sd_reg_mr()` allocated a dma-heap buffer, memcpy'd the caller's bytes into it and returned the copy's address as `mr->addr`, so later writes to the caller's own buffer never reached the fabric. It set `lkey = dma_buf_fd` and `rkey = fd ^ 0xA5A55A5A`, a reversible function of a small, reused integer. It ignored `remote_register_buf_attr()`'s return value and left two `calloc` results unchecked.

8.2 Send path

[R-8.3-001] A send SHALL take exactly one path. `sd_post_send()` SHALL either enqueue to the hardware ring **or** transmit over the socket, never both. v2.0 enqueued the work request and then also called `sd_send_via_socket()`, so a request could be submitted twice.

[R-8.3-002] A socket transmission SHALL serialise the work request into the Profile B packet of Section 5.1. A queued struct `uaf_wr` contains pointers and is not a packet.

[R-8.3-003] Doorbell ordering follows Section 9.4. The doorbell value SHALL be the new ring tail, not the queue-pair number: a device cannot infer a tail from an identifier.

```

static int sd_post_send(uaf_qp_t *qp, struct uaf_wr *wr)
{
    struct uaf_s_qp *q = uaf_s_qp(qp);
    int rc = sd_validate_wr(q, wr); /* num_sge, inline, opcode */
    if (rc != UAF_OK) return rc;

    if (q->mode == UAF_S_MODE_SOCKET) /* [R-8.3-001] one path */
        return sd_send_serialized(q, wr);

    uint32_t slot;
    struct uaf_wr *e = uaf_ring_produce(&q->sq, &slot);
    if (!e) return UAF_ERR_BUSY;

    *e = *wr; /* [R-4.4-002] copy the WR */
    memcpy(q->sge_pool + slot * q->max_sge, wr->sge,
          wr->num_sge * sizeof(struct uaf_sge)); /* and the SGE array */
    e->sge = q->sge_pool + slot * q->max_sge;
    uaf_ring_commit(&q->sq);
}

```

```

sd_dma_sync_for_device(q); /* [R-8.2-003] */
uaf_mmio_write32(q->doorbell, q->sq.tail); /* [R-8.3-003], [R-9.4-001] */
return UAF_OK;
}

```

9 Platform Backend: UAF-D (Dragonfly)

Listings in this section are INFORMATIVE ([R-1.4-003]).

[R-9.1-001] UAF-D SHALL declare UAF_PROFILE_UAFR for node-to-node traffic. Its intra-host path is CXL.mem load/store through the ring of Section 9.2 and carries no packet header.

9.1 Memory registration

[R-9.1-002] MAP_SYNC SHALL be requested with MAP_SHARED_VALIDATE, never with MAP_SHARED. Under MAP_SHARED the flag is not validated, and the outcome is filesystem-dependent: it may be **silently ignored**, returning a successful mapping with none of the synchronous-fault guarantee the caller asked for, or it may be refused with EOPNOTSUPP. Both are unacceptable for a durability-sensitive mapping, and the silent case is the more dangerous of the two. MAP_SHARED_VALIDATE is the spelling the kernel validates, and it returns EOPNOTSUPP when the file has no DAX sync support.

v2.1 asserted that v2.0's MAP_SHARED | MAP_SYNC fails on every kernel. That over-generalised from one filesystem. Measured on Linux 6.18: on tmpfs (/dev/shm) MAP_SHARED | MAP_SYNC **succeeded**, with MAP_SYNC dropped; on another filesystem it returned EOPNOTSUPP. MAP_SHARED_VALIDATE | MAP_SYNC returned EOPNOTSUPP on both, correctly, neither being DAX. The conclusion is unchanged and the reason is sharper: v2.0 risked losing the persistence guarantee without an error.

[R-9.1-003] If MAP_SHARED_VALIDATE | MAP_SYNC returns EOPNOTSUPP the implementation MAY fall back to MAP_SHARED and SHALL then clear UAF_CAP_SYNC_FAULT.

[R-9.1-004] Device-DAX alignment SHALL be honoured. A device-DAX instance commonly requires 2 MiB or 1 GiB alignment for both the mapping length and the file offset; a request that does not satisfy it SHALL fail with UAF_ERR_INVALID.

[R-9.1-005] uaf_reg_mr() on UAF-D SHALL succeed only for an address already inside a registered CXL window. Otherwise it SHALL return UAF_ERR_NOT_SUPPORTED, and the backend SHALL clear UAF_CAP_REG_ANY_MEM. v2.0 passed the caller's address to mmap as a hint without MAP_FIXED, which creates a new mapping somewhere else and registers that instead.

```

struct uaf_d_mr { uaf_mr_t pub; int dax_fd; uint64_t cxl_base; };

static int df_alloc_mr(uaf_domain_t *pd, size_t len, uint64_t flags,
                      uaf_mr_t **out)
{
    uint64_t align = df_dax_alignment(pd); /* [R-9.1-004] */
    if (align == 0u || (len & (align - 1u))) return UAF_ERR_INVALID;

    int fd = open(pd->dax_path, O_RDWR | O_CLOEXEC);
    if (fd < 0) return UAF_ERR_NODEV;

    uint64_t off;
    int rc = df_alloc_cxl_window(pd, len, align, &off); /* checked */
}

```

```

if (rc != UAF_OK) { close(fd); return rc; }

int mflags = MAP_SHARED_VALIDATE | MAP_SYNC; /* [R-9.1-002] */
void *va = mmap(NULL, len, PROT_READ | PROT_WRITE, mflags, fd,
                (off_t)off);
if (va == MAP_FAILED && errno == EOPNOTSUPP) {
    va = mmap(NULL, len, PROT_READ | PROT_WRITE, MAP_SHARED, fd,
              (off_t)off);
    if (va != MAP_FAILED) pd->caps &= ~UAF_CAP_SYNC_FAULT; /* [R-9.1-003] */
}
if (va == MAP_FAILED) {
    df_free_cxl_window(pd, off, len); close(fd); return UAF_ERR_NOMEM;
}

uint32_t lkey = 0, rkey = 0;
/* Unbound until uaf_connect() calls uaf_key_bind_qp() ([R-11.2-003]). */
rc = uaf_key_register(&pd->keys, (uint64_t)(uintptr_t)va, len, flags,
                    0u, &lkey, &rkey); /* [R-11.2-001] */
if (rc != UAF_OK) {
    munmap(va, len); df_free_cxl_window(pd, off, len); close(fd);
    return rc;
}
struct uaf_d_mr *m = calloc(1, sizeof(*m));
if (!m) {
    uaf_key_deregister(&pd->keys, rkey); munmap(va, len);
    df_free_cxl_window(pd, off, len); close(fd); return UAF_ERR_NOMEM;
}
m->dax_fd = fd; m->cxl_base = off;
m->pub.struct_size = (uint32_t)sizeof(uaf_mr_t);
m->pub.pd = pd; m->pub.addr = va; m->pub.length = len;
m->pub.lkey = lkey; m->pub.rkey = rkey; m->pub.flags = flags;
*out = &m->pub;
return UAF_OK;
}

```

9.2 Send ring over CXL.mem

[R-9.2-001] The send ring index SHALL wrap on the **send ring's own depth**, which SHALL be a power of two. v2.0 wrapped with % UAF_CQ_DEPTH_DEFAULT (1024) a ring whose default depth was UAF_SQ_DEPTH_DEFAULT (256), writing up to 49152 bytes past the end for a 64-byte work request.

[R-9.2-002] Producer and consumer indices SHALL follow Section 10.3: the producer advances tail. v2.0's send path advanced head while its storage path advanced tail, in adjacent sections of one backend.

[R-9.2-003] UAF_DOORBELL_SQ and every other doorbell selector SHALL be defined in a public header. v2.0 used the identifier and defined it nowhere.

```

#define UAF_DOORBELL_SQ 0u
#define UAF_DOORBELL_RQ 1u
#define UAF_DOORBELL_CQ 2u

static int df_post_send(uaf_qp_t *qp, struct uaf_wr *wr)
{
    struct uaf_d_qp *d = uaf_d_qp(qp);
    int rc = df_validate_wr(d, wr);
}

```

```

if (rc != UAF_OK) return rc;

uint32_t slot;
struct uaf_wr *e = uaf_ring_produce(&d->sq, &slot); /* [R-9.2-001] */
if (!e) return UAF_ERR_BUSY;

*e = *wr;
memcpy(d->sge_pool + slot * d->max_sge, wr->sge,
        wr->num_sge * sizeof(struct uaf_sge));
e->sge = d->sge_pool + slot * d->max_sge;
uaf_ring_commit(&d->sq); /* [R-9.2-002] */

df_ring_doorbell(qp, UAF_DOORBELL_SQ, d->sq.tail);
return UAF_OK;
}

```

9.3 DST over a CXL-mapped NVMe BAR

```

static int df_storage_submit(uaf_sq_t *sq, struct uaf_storage_cmd *cmd)
{
    struct uaf_d_sq *s = uaf_d_sq(sq);
    if (cmd->struct_size != sizeof(*cmd)) return UAF_ERR_INVALID;

    int opc = uaf_dst_nvme_opcode(cmd->opcode); /* [R-6.2-001] */
    if (opc < 0) return opc;

    uint16_t nlb0 = 0;
    if (opc != UAF_NVME_OPC_FLUSH) { /* [R-6.2-002] */
        int rc = uaf_dst_nlb_to_cdw12(cmd->nlb, &nlb0);
        if (rc != UAF_OK) return rc;
        /* Ordered so the subtraction cannot wrap: v2.1's sample computed
         * buf_len - buf_offset first, which underflows to a huge value when
         * buf_offset > buf_len and lets an illegal command through. */
        if (cmd->buf_offset > cmd->buf_len ||
            cmd->buf_len - cmd->buf_offset <
            (uint64_t)cmd->nlb * s->block_size) return UAF_ERR_INVALID;
    }

    uint16_t cid;
    int rc = uaf_cid_alloc(&s->cids, cmd->wr_id, &cid); /* [R-6.5-001] */
    if (rc != UAF_OK) return rc;

    uint32_t slot;
    struct nvme_sq_entry *e = uaf_ring_produce(&s->ring, &slot);
    if (!e) { uaf_cid_release(&s->cids, cid, NULL); return UAF_ERR_BUSY; }

    memset(e, 0, sizeof(*e));
    e->opcode = (uint8_t)opc;
    e->cmd_id = cid;
    e->nsid = s->nsid;
    if (opc != UAF_NVME_OPC_FLUSH) {
        uint64_t iova = s->iova_base + cmd->buf_offset;
        enum uaf_dst_dpnr kind;
        rc = uaf_dst_dpnr_kind(iova, (uint64_t)cmd->nlb * s->block_size,
                               s->page_size, s->sgl_supported, &kind);
        if (rc != UAF_OK) { uaf_cid_release(&s->cids, cid, NULL); return rc; }
    }
}

```

```

    df_fill_dptr(e, kind, iova, cmd);                /* [R-6.2-003] */
    e->cdw10 = (uint32_t)(cmd->lba & 0xFFFFFFFFFull);
    e->cdw11 = (uint32_t)(cmd->lba >> 32);
    e->cdw12 = nlb0;
}
uaf_ring_commit(&s->ring);

df_sq_clean_to_memory(s);                          /* [R-6.6-001] */
uaf_mmio_write32(s->sq_tdbl, s->ring.tail);        /* [R-9.4-001] */
return UAF_OK;
}

```

9.4 Device-visible ordering

[R-9.4-001] A doorbell write SHALL be preceded by a barrier that orders prior descriptor stores against a store to device memory. `atomic_thread_fence(memory_order_seq_cst)` is **not** such a barrier: on AArch64 it compiles to `dmb ish`, which orders inner-shareable cacheable accesses and says nothing about Device-nGnRE. v2.0's "doorbell ordering fix" used exactly that fence on both software backends.

[R-9.4-002] The doorbell page SHALL be mapped as device memory (Device-nGnRE on AArch64, uncached or write-combining on x86).

[R-9.4-003] Producer and consumer indices in a ring shared with a device SHALL both be accessed atomically. v2.0's storage path stored `sq_tail` as a plain `uint32_t` while loading `sq_head` atomically.

```

static inline void uaf_dma_wmb(void)
{
    #if defined(__aarch64__) || defined(__arm__)
        __asm__ __volatile__ ("dsb st" ::: "memory");
    #elif defined(__x86_64__) || defined(__i386__)
        __asm__ __volatile__ ("sfence" ::: "memory");
    #else
        atomic_thread_fence(memory_order_seq_cst);
    #endif
}

static inline void uaf_mmio_write32(volatile void *addr, uint32_t v)
{
    uaf_dma_wmb();
    *(volatile uint32_t *)addr = v;
}

```

9.5 Intra-host binding and the ring consumer

The Profile B connection manager of Section 5.5 does not apply to a path that carries no packet. v2.1 specified the producer side of the CXL.mem ring and left three things unstated: who consumes it, how `remote_va` becomes a CXL offset, and how `uaf_connect()` binds two windows. Without them Section 9.2 could be implemented and not used.

[R-9.5-001] **The consumer.** The accelerator-side agent on the C1000 SHALL consume the ring. It polls `tail`, reads the descriptor at `head & mask`, performs the transfer, writes the completion, and advances `head`. The doorbell of [R-9.2-003] is a hint that lets the agent avoid a busy poll; correctness SHALL NOT depend on the doorbell being observed, because a lost doorbell must not lose a descriptor.

[R-9.5-002] **Address form.** On UAF-D `uaf_wr.remote_addr` SHALL be a byte offset **relative to the base of the peer's CXL window**, not a host virtual address. The agent computes the host-visible address as `cxl_base + remote_addr`, and SHALL reject with `UAF_ERR_MR_FAULT` any request for which

```
remote_addr > cxl_size || L > cxl_size - remote_addr
```

where `L` is the sum of the work request's SGE lengths. Both terms are required, and in this order.

[R-9.5-010] Every bounds check in this specification SHALL be written so that it cannot be defeated by integer overflow. The natural spelling `offset + length > size` is **not** such a check: the sum wraps modulo 2^{64} , so an offset of $2^{64} - 8$ with a length of 16 yields 8 and passes. v2.1.3 stated [R-9.5-002] in that form.

```
/* The single spelling used throughout. len == 0 is within any size. */
static inline int uaf_range_within(uint64_t off, uint64_t len, uint64_t size)
{
    if (len == 0u) return 1;
    if (off > size) return 0;
    return len <= size - off;
}
```

Conformance cases C-5 and C-12 both drive the wrapping case.

[R-9.5-003] **Binding.** A queue pair on this path SHALL carry `path = UAF_PATH_CXL` in `uaf_qp_init_attr` and in the `uaf_conn_info` exchanged with the peer. `uaf_connect()` on UAF-D SHALL NOT send CM packets. It takes a struct `uaf_conn_info` whose `cxl_base`, `cxl_size` and `rkey` were obtained out of band, validates that the window lies inside a region this domain may reach, installs the binding — including the queue-pair component of every `rkey` it will accept, per [R-11.2-003] — and moves the queue pair `INIT` → `RTR` → `RTS` without any wire exchange. The required RTR mask is the **base mask of [R-10.1-001] plus the UAF_PATH_CXL row of [R-10.1-008]**, that is `STATE | RQ_PSN | DEST_QPN | MAX_DEST_RD_AT | AV_CXL`. Section 10.1 is authoritative; v2.1.1 restated the mask here and the two disagreed, and v2.1.2's wording could be read as the CXL row alone.

[R-9.5-004] **Authentication.** Because there is no CM exchange, the authenticator of [R-5.5-006] does not apply. The binding is authenticated by the fact that both windows are mapped by the same host kernel under the IOMMU tables of [R-11.1-004]. `rkey` validation ([R-11.2-003]) and the bounds check of [R-9.5-002] apply unchanged, and the bounds check is the control that matters here.

The throttles of [R-11.2-004] and [R-11.2-008] do **not** apply to this path. [R-11.2-004] counts failures on packets that passed a link authenticator and [R-11.2-008] counts them per source address; this path has no packets and no source address, so neither counter can advance on it. An invalid `rkey` here is a local programming error, not a remote probe: the only writer is an agent the host itself programmed. v2.1.3 cited [R-11.2-004] on this path, which described a control that cannot engage.

[R-9.5-005] **Completions.** The agent SHALL write a struct `uaf_rmt_cqe` into the completion ring. The host SHALL convert an accepted entry into a struct `uaf_wc` before delivering it to the caller.

```
struct uaf_rmt_cqe {
    uint64_t wr_id;
    int32_t status;           /* enum uaf_error */
    uint32_t byte_len;
    uint32_t imm_data;
    uint32_t wc_flags;       /* UAF_WC_* */
}
```

```

uint8_t reserved0[7];      /* MUST be zero */
uint8_t phase;             /* bit 0 is the phase tag */
};                          /* 32 bytes, phase at offset 31 */

```

[R-9.5-006] The completion ring SHALL be a separate array of struct uaf_rmt_cqe in the same CXL window as the send ring, with the same depth as the send ring, at the offset given by [R-9.5-009]. It SHALL be 32-byte aligned and zero-filled at creation.

[R-9.5-007] Ordering SHALL be the **phase byte**, not an acquire on head. The producer writes bytes 0..30, issues a release, then stores phase; the consumer acquire-loads phase first and reads the body only on a match. An acquire on a shared head would make the host trust an index the agent advances, whereas the phase tag makes each entry self-describing. The ring inherits the **whole** phase discipline of the DST ring — [R-6.3-002], [R-6.3-003] and [R-6.3-005] — and in particular the **initial expected phase is 1** and it **flips on every wrap**.

v2.1.2 cited only the producer release and the consumer acquire. Without the initial value, a consumer that began expecting phase 0 would accept every entry of a zero-filled ring as a completion with status UAF_OK, which is the precise failure the phase tag exists to prevent. Conformance case C-12 asserts the initial value, so the suite and the requirement text had disagreed about whether the behaviour was specified at all.

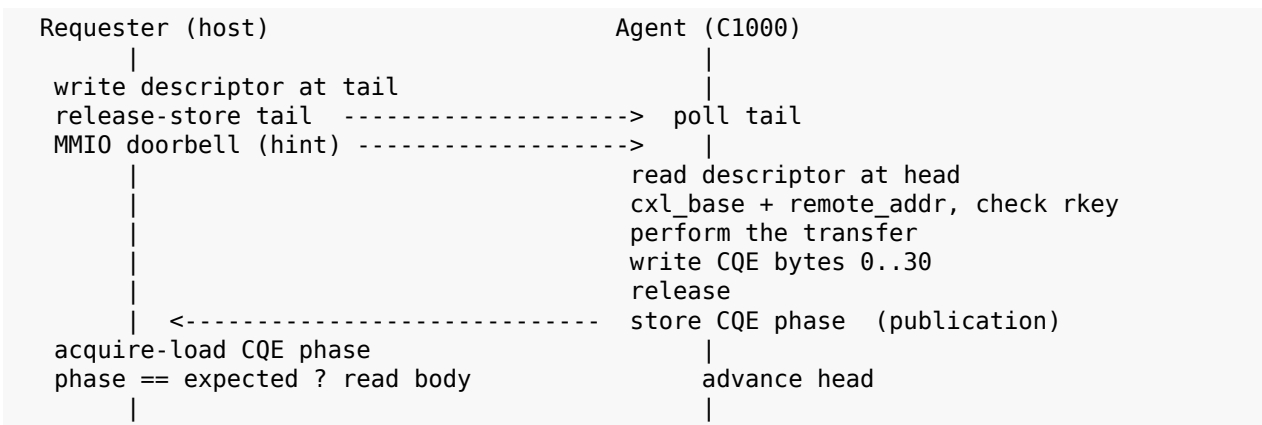
[R-9.5-009] The completion ring SHALL immediately follow the send ring in the window, at

$$cq_offset = \left\lceil \frac{sq_offset + depth \times sizeof(struct\ uaf_wr)}{32} \right\rceil \times 32$$

that is, the send ring’s base plus depth × 64 bytes, rounded up to the 32-byte alignment of [R-9.5-006]. Both ends compute it; no field carries it. v2.1.2 said the offset was “recorded at queue-pair creation” and put it in no struct, so the two ends had nothing to agree on.

[R-9.5-008] The host SHALL reject an entry whose reserved0 bytes are non-zero with UAF_ERR_PROTO.

v2.1.1 told the agent to release-store “the completion” and the host to acquire-load it, named no object, and offered both ordering mechanisms without choosing. uaf_wc is 40 bytes, host-side, and has no phase byte.



Publication is the phase-byte store, not the advance of head. The agent advances head to reclaim the descriptor slot; that is bookkeeping between the agent and itself, and the host never reads it. v2.1.2’s diagram showed the host acquire-loading “the completion” after the agent advanced head, which reads as the mechanism [R-9.5-007] rejects.

10 State Machines, Memory, and Rings

10.1 Queue-pair state machine

[R-10.1-001] The legal transitions are exactly these. Every other transition SHALL fail with UAF_ERR_QP_STATE.

From	To	Trigger	Required attribute mask
RESET	INIT	modify_qp(INIT)	STATE, PORT
INIT	RTR	modify_qp(RTR)	STATE, RQ_PSN, DEST_QPN, MAX_DEST_RD_AT, plus the per-path columns below
RTR	RTS	modify_qp(RTS)	STATE, SQ_PSN, MAX_RD_ATOMIC, RETRY, RNR_RETRY, TIMEOUT
RTS	SQD	modify_qp(SQD)	STATE
SQD	RTS	modify_qp(RTS)	STATE
any	ERR	fault, or modify_qp(ERR)	STATE
any	RESET	modify_qp(RESET)	STATE

[R-10.1-008] The additional RTR requirements are **per path**, not per profile:

uaf_path	Additional RTR mask	MTU required
UAF_PATH_IBV	AV_GID, AV_LID	Yes, PATH_MTU
UAF_PATH_UDP	AV_UDP	Yes, PATH_MTU
UAF_PATH_CXL	AV_CXL	No — the path carries no packet

[R-10.1-002] A CXL or UDP path has neither a GID nor a LID, and v2.0's RTR rule demanded both from every backend. v2.1.1 then made the rule profile-dependent, which still failed for UAF-D: it declares Profile B, so [R-10.1-002] required PATH_MTU and AV_UDP of a queue pair that [R-9.5-003] said needed neither. The path discriminant resolves it, and Section 10.1 is authoritative.

[R-10.1-003] rq_psn is REQUIRED at RTR. v2.0's table never asked for it.

[R-10.1-004] UAF_QPS_SQE is entered by the implementation on a send-queue error and is left only through RESET or ERR. v2.0 declared the state in its enum and gave it no row in the table, so it was unreachable and undocumented.

[R-10.1-005] A work request already in flight when a fault occurs SHALL complete with the **originating** error — UAF_ERR_TIMEOUT, UAF_ERR_CRC, UAF_ERR_RKEY and so on. Only requests flushed *after* the transition complete with UAF_ERR_QP_STATE. v2.0 said every fault flushed with UAF_ERR_QP_STATE, which discarded the very status the rest of the document takes care to produce.

10.2 Key encoding

[R-10.2-001] A backend that owns its own key space — UAF-S and UAF-D — SHALL draw `lkey` and `rkey` from the software key table of Section 11.2, bound to a domain, a queue pair, a base, a length and a generation.

Backend	<code>lkey</code>	<code>rkey</code>	Address carrier
UAF-N	<code>ibv_mr->lkey</code>	<code>ibv_mr->rkey</code>	64-bit GPU or host virtual address
UAF-S	CSPRNG, key table	CSPRNG, key table, distinct from <code>lkey</code>	64-bit virtual address
UAF-D	CSPRNG, key table	CSPRNG, key table, distinct from <code>lkey</code>	64-bit CXL window offset

[R-10.2-002] On UAF-N the adapter owns the key space. The verbs keys are authoritative, the software table is not used, and a UAF-N implementation SHALL NOT synthesise its own keys. [R-11.2-001] through [R-11.2-003] and [R-11.2-005] apply to the software key table only and therefore do not apply to UAF-N; [R-11.2-004] and [R-11.2-006] apply to every backend.

[R-10.2-003] A UAF-N implementation MAY report the same 32-bit value for `lkey` and `rkey`, because `ibv_reg_mr()` commonly returns one value for both.

v2.1 was unsatisfiable here. [R-10.2-001] required every backend to use the CSPRNG table, [R-10.2-002] forbade UAF-N from synthesising keys, and [R-11.2-002] required `lkey != rkey` — which verbs violates. A conforming UAF-N implementation could not exist.

v2.0's table described the UAF-D `rkey` as “32-bit random” while its code derived it from the CXL offset, length and flags, and described the UAF-S `lkey` as a descriptor number with the `rkey` a salted copy of it.

10.3 Ring discipline

[R-10.3-001] In every ring in this specification the **producer advances tail** and the **consumer advances head**, following NVMe.

[R-10.3-002] Every ring depth SHALL be a power of two and index wrap SHALL be a mask.

[R-10.3-003] The modulus SHALL be the depth of the ring being indexed. A constant naming a different queue MUST NOT appear in an index computation.

[R-10.3-004] Occupancy SHALL be computed as `tail - head` in unsigned arithmetic, so a counter wrap at 2^{32} is well defined.

[R-10.3-005] Rings are single-producer, single-consumer ([R-4.9-001]).

11 Security

11.1 Threat model and mitigations

[R-11.1-001] The deployment model SHALL be stated in a conformance claim. Two are recognised: a **trusted fabric**, where every endpoint is administratively controlled, and an **untrusted fabric**, where it is not.

[R-11.1-002] On an untrusted fabric, link or network-layer authenticated encryption — PSP, IPsec ESP or MACsec — is REQUIRED, not optional. v2.0 listed them as optional with no key exchange, no SPI and no “required on the wire” rule, which left cleartext remote DMA behind an unauthenticated handshake as the default.

Threat	Mitigation	Requirement
Unauthorised remote write or read	CSPRNG rkey, bound to domain, QP, base, length and generation; bounds and access-flag check; IOMMU or SMMU	[R-11.2-001]..[R-11.2-005]
rkey brute force	Consecutive authenticated-failure limit, then validation is rate-limited. Queue-pair state is never changed	[R-11.2-004]
Off-path denial of service via forged rkey	Unauthenticated failures are not counted, so they cannot engage the throttle	[R-11.2-004], [R-5.7-001]
rkey brute force on a fabric with no link authenticator	Throttled per source address, so the counter still advances and one source cannot slow another	[R-11.2-008]
Stale rkey after deregistration	Generation counter invalidates it	[R-11.2-005]
Payload corruption	hdr_crc32c and data_crc32c	[R-5.2-002], [R-6.3-002]
Packet injection or modification	PSP, IPsec ESP or MACsec. CRC32C does not mitigate this	[R-11.1-002], [R-11.1-003]
Connection hijack, CM spoofing	HMAC-SHA256 authenticator on every CM packet	[R-5.5-006]
CM replay	Nonce cache and timestamp window	[R-5.5-007]
Connection teardown by an off-path host	A malformed packet never moves a QP to ERR	[R-5.7-001]
Malicious DMA from a peripheral	Per-domain IOMMU or SMMU stage-1 tables	[R-11.1-004]
Physical CXL bus snooping	CXL IDE on UAF-D links	[R-11.1-005]
Confused deputy via a descriptor	No descriptor in any off-node structure	[R-5.5-005]

[R-11.1-003] CRC32C is an error-detecting code, not an integrity or authentication mechanism. An attacker who alters a packet recomputes it for free. v2.0’s threat matrix named “Header CRC32C” as the mitigation for packet injection, which is a category error, and covered only header bytes 0..39 — leaving the payload, the CRC field itself and the reserved area outside any check.

[R-11.1-004] Every domain SHALL have its own IOMMU or SMMU stage-1 translation tables. A region SHALL NOT be reachable from a domain that did not register it.

[R-11.1-005] A UAF-D deployment carrying data outside a physically secured enclosure SHALL enable CXL Integrity and Data Encryption (IDE).

11.2 Key management

Scope. [R-11.2-001] through [R-11.2-003] and [R-11.2-005] govern the **software key table**, used by UAF-S and UAF-D. On UAF-N the adapter owns the key space and the verbs keys are authoritative ([R-10.2-002]).

[R-11.2-001] lkey and rkey SHALL be generated by a CSPRNG. Neither SHALL be derived from an address, an offset, a length, a file descriptor, a counter, or any other value an attacker can guess or observe.

[R-11.2-002] lkey and rkey SHALL be distinct and non-zero. v2.0's UAF-D code set them equal, so disclosure of a local key conferred remote write. This requirement does not extend to UAF-N ([R-10.2-003]).

[R-11.2-003] An rkey SHALL be bound to (domain, queue pair, base, length). A request presenting a valid rkey on a queue pair it was not issued for SHALL fail with UAF_ERR_RKEY. The queue-pair component SHALL be installed at **connect** time, because a queue pair does not exist when memory is registered, and an rkey that has not yet been bound SHALL NOT validate. v2.1.1's informative samples registered with a queue-pair number of zero and never bound, so this requirement described a state nothing reached.

[R-11.2-004] After UAF_RKEY_FAIL_MAX consecutive validation failures — counted **only for packets that passed the link authenticator** — a responder SHALL throttle rkey validation on that queue pair to one attempt per UAF_RKEY_THROTTLE_NS (1 ms). It MUST NOT change queue-pair state. A 32-bit key space is otherwise exhaustible in seconds: at 400 Gbps a 64-byte probe rate of order 10^9 per second covers 2^{32} keys in about four seconds, whereas at one attempt per millisecond the same search takes over a century.

v2.1 moved the queue pair to UAF_QPS_ERR here, which contradicted [R-5.7-001] and handed any host able to reach the UDP port a connection kill in sixteen datagrams. Counting only authenticated failures also stops an off-path attacker from driving the responder into throttling itself. Conformance case C-5 asserts that unauthenticated failures never engage this counter. **Where a link authenticator is configured**, that is the whole rule; where none is, [R-11.2-008] supplies the per-source counter instead, and the sentence above about unauthenticated failures does not leave the key space unprotected.

[R-11.2-005] Deregistration SHALL increment the domain's generation counter so a stale rkey can never match a later registration.

[R-11.2-006] A successful validation SHALL reset the failure counter.

[R-11.2-008] **Where no link authenticator is configured** — permitted on a trusted fabric by [R-11.1-002] — a failure cannot be attributed to an authenticated peer, so [R-11.2-004]'s counter never advances and the 32-bit key space is scannable in seconds again, in the one deployment where the rkey is the only secret. A responder in that configuration SHALL throttle rkey validation **per source address** instead, on the same threshold and interval. Queue-pair state is still never changed, so the off-path kill of [R-5.7-001] stays closed, and one abusive source cannot slow validation for any other peer. v2.1.3 offered an alternative here — requiring the fabric key to cover data packets — with no byte layout for the result, making it a normative option that nothing could implement interoperably. It is withdrawn. A revision that wants it SHALL define the packet format first.

[R-11.2-007] **Key management is an external dependency.** The pre-shared fabric key of [R-5.5-006], and any PSP, IPsec or MACsec keys required by [R-11.1-002], SHALL be supplied by a key-management service outside this specification and SHALL be rotated on that service's schedule. An implementation SHALL obtain them from a platform key store — a TPM-sealed blob, a kernel keyring, or an operator-provided KMS client — and SHALL NOT embed a key in a binary, a configuration file in the source tree, or an environment variable. uaf_init() SHALL fail with UAF_ERR_AUTH if a required key is unavailable. v2.1 made the CM authenticator mandatory and named no source for its key.

Conformance case C-5 exercises every clause above, including the brute-force limit and the stale-key case.

12 Performance and Conformance

12.1 Measurement method

[R-12.1-001] A performance figure in this specification is a **target**, not a measurement. A conformance claim SHALL report measured values obtained by the method below, and SHALL state the hardware, firmware and driver versions. v2.0 published a table with no percentile, no queue depth, no message size for the bandwidth rows, no statement of CPU versus GPU submission, and no pass/fail rule.

[R-12.1-002] The method SHALL be:

Parameter	Definition
Harness	uaf_perf from the conformance tree, one process per node
Latency metric	One-way, half of the round trip of a ping-pong, reported at p50, p99 and p99.9
Latency message sizes	64 B and 4096 B, reported separately
Latency queue depth	1
Bandwidth metric	Unidirectional goodput, payload bytes only, excluding the 64-byte header
Bandwidth message size	1 MiB
Bandwidth queue depth	64 outstanding messages per queue pair
IOPS metric	4 KiB random reads, queue depth 128 per queue, reported per device and per node
Warm-up	10 s discarded
Duration	60 s minimum
Scope	Stated as per queue pair, per device, or per node
Pass criterion	p99 latency within 20 % of target, bandwidth and IOPS within 10 %, zero errors and zero retransmissions on an idle fabric

12.2 Targets

[R-12.2-001] These targets replace v2.0's table. Each is accompanied by the ceiling it is derived from, so a reader can check it.

12.2.1 RMT latency (one-way, p50)

Path	64 B	4 KiB	Basis
UAF-N, IB RC	1.5 μ s	2.0 μ s	ConnectX-7/8 small-message RC; 4 KiB adds 82 ns of wire time at 400 Gbps
UAF-N, RoCEv2 RC	2.5 μ s	3.0 μ s	As above with Ethernet encapsulation
UAF-S, PCIe P2P or DSP	8 μ s	10 μ s	On-package path, no network stack

Path	64 B	4 KiB	Basis
UAF-S, UAF-RMT over UDP	20 μ s	25 μ s	Software UDP on an X Elite class core. v2.0 quoted 25 μ s without distinguishing this from the P2P path, and the platform has no RoCE offload
UAF-D, direct CXL.mem store	0.4 μ s	0.5 μ s	CXL 3.1 load-to-use of order 100–250 ns plus 32 ns to move 4 KiB at 128 GB/s
UAF-D, software ring (Section 9.2)	1.8 μ s	2.0 μ s	Adds a work-request copy, an MMIO doorbell and a responder poll

v2.0 listed 0.3 μ s for UAF-D without saying which path. The direct load/store path is close to that figure; the path the specification actually defines in Section 9.2 includes a ring, a work-request copy and a doorbell, and cannot reach it.

12.2.2 RMT bandwidth (unidirectional goodput, 1 MiB messages)

Path	Target	Link ceiling	Basis
UAF-N, ConnectX-7	380 Gbps	400 Gbps	95 % of line rate
UAF-N, ConnectX-8	760 Gbps	800 Gbps	95 % of line rate
UAF-S, PCIe P2P	64 Gbps	126 Gbps (PCIe 4.0 x8)	8 GB/s on the local path
UAF-S, UDP	25 Gbps	—	Software transport, multiple cores
UAF-D, CXL 3.1 x16	900 Gbps per direction	1024 Gbps raw, $\approx$ 940–970 Gbps usable, per direction	64 GT/s $\times$ 16 lanes; FLIT and protocol overhead removes roughly 6–8 %. 2048 Gbps is the duplex aggregate
UAF-D, CXL 4.0 x16 (OPTIONAL)	1800 Gbps per direction	2048 Gbps raw, $\approx$ 1880–1940 Gbps usable	128 GT/s $\times$ 16 lanes, requires CXL 4.0

[R-12.2-004] A bandwidth ceiling SHALL distinguish the **raw** line rate from **usable** goodput. The raw per-direction rate of a CXL 3.1 x16 link is the product $64 \text{ GT/s} \times 16 = 1024 \text{ Gbps}$; FLIT framing, CRC, FEC and protocol overhead remove roughly 6–8 %, so the usable ceiling is about 940–970 Gbps. v2.1 quoted a 950 Gbps target against the raw figure and called it 93 % of line rate, which understated how close to the usable ceiling it sat. The target is now 900 Gbps and the ceiling is stated as a band, pending measurement under OI-7.

[R-12.2-002] A bandwidth figure SHALL state whether it is per direction or a duplex aggregate. v2.0's "2,000 Gbps peak unidirectional" for a CXL 3.1 link was the duplex aggregate of a PCIe 7.0 PHY quoted against a CXL 3.1 specification: two errors compounded.

12.2.3 DST throughput (4 KiB random read, QD 128)

Path	Per device	Per node	Basis
UAF-N	2.0 M IOPS per GPU	16 M IOPS, 8 GPUs	PCIe 5.0 x16 carries 13.4 M IOPS of 4 KiB traffic at 55 GB/s; 2.0 M is 15 % of that ceiling and is media-bound. Requires at least 8 PCIe 5.0 NVMe devices
UAF-S	1.2 M IOPS	1.2 M IOPS	A PCIe 4.0 x4 client NVMe device
UAF-D	3.0 M IOPS	12 M IOPS, 4 devices	CXL-mapped NVMe; media-bound, not link-bound

[R-12.2-003] v2.0's "200 M+ IOPS" for UAF-N is withdrawn. At 4 KiB it implies $200 \times 10^6 \times 4096 = 819$ GB/s of storage traffic, roughly fifteen times a single PCIe 5.0 x16 link and beyond any published GPUDirect Storage result on one node. Likewise 50 M IOPS for UAF-D implies 205 GB/s from the storage media behind one accelerator.

12.3 Conformance suite

[R-12.3-001] An implementation SHALL pass every case below. The suite is in conformance/ and is built and run by conformance/run_conformance.sh, which configures with -DUAF_BUILD_TESTS=0N and runs ctest. v2.0's Section 12 declared UAF_BUILD_TESTS and then ended, with no test of any kind.

Case	Name	Covers	Requirements
C-1	abi	Struct sizes, offsets, CQE alignment and phase position, MTU accounting, no size_t or enum field	R-4.1-001..005, R-6.3-001
C-2	crc32c	RFC 3720 vectors and the golden header of Section 5.8	R-5.8-001
C-3	wire	Encode and decode round trip, big-endian order, bad magic, bad version, header CRC failure, non-zero reserved bytes and flags, UAF_WR_RECV rejected, CM and ACK/NAK opcodes, framing and segment accounting, payload CRC, atomic lengths, misaligned atomic, PSN wrap	R-5.2-002, R-5.3-001..007, R-5.4-001..004, R-5.6-001..003, R-5.7-001

Case	Name	Covers	Requirements
C-4	qp_state	Legal and illegal transitions, SQE reachability, the three per-path RTR columns including a CXL path with no MTU, attribute-mask enforcement, attr->qp_state agreement, in-flight status preservation	R-4.5-003, R-4.5-004, R-10.1-001..008
C-5	rkey	Key generation, distinctness, binding installed at connect, an unbound key refused, bounds including the overflow case, access flags, authenticated brute-force throttle, unauthenticated failures ignored, per-source throttling with no authenticator, generation invalidation	R-9.5-010, R-11.2-001..008
C-6	ring	Power-of-two depth, full ring, wrap inside bounds across $4 \times$ depth, counter wrap at 2^{32} , independence of the two default depths	R-9.2-001, R-10.3-001..004
C-7	cqe_phase	Zeroed ring yields zero completions, batch bound by max, phase flip on wrap, stale entries not re-reported, failure status preserved, latency clamped, the full 64-bit wr_id survives uaf_storage_poll(), an unknown cmd_id is a protocol error	R-4.7-002, R-6.3-001..006, R-6.5-001..003
C-8	dst_map	Opcode map including flush, NLB 0 rejected, NLB 1 and 65536, PRP1/PRP2/PRP-list/SGL selection, cmd_id allocation and 64-bit wr_id round trip, table exhaustion, each NVMe status code point pinned by value, and an exhaustive sweep for silent success	R-6.2-001..003, R-6.5-001..003, R-6.7-001..002

Case	Name	Covers	Requirements
C-9	conn_wire	Connection record round trip, big-endian layout, host memcpy is not the wire form, no descriptor on the wire, reserved bytes, ambiguous profile, MTU floor, the path byte and its consistency with the profile, and that the CXL path is exempt from the MTU floor	R-5.5-002..005, R-5.5-012
C-10	udp_loopback	An end-to-end multi-segment RDMA WRITE over UDP: segmentation, transmission, reassembly, one completion on LAST, byte-exact result, corrupted payload dropped without writing, wrong rkey refused, write past the region refused	R-4.11-002, R-5.3-003..005, R-5.7-001, R-11.2-003
C-11	cm	CM body length per opcode including a 32-byte RTU, the MAC input for both body sizes, replay keyed on the header qp_id, canonical endpoint identity from source fields, strict total order, and exactly one active side for every distinguishing field	R-5.5-001, R-5.5-006, R-5.5-007, R-5.5-009, R-5.5-013
C-12	rmt_cq	The intra-host RMT completion ring: an initial expected phase of 1, phase-tag polling, a zeroed ring yields none, conversion to uaf_wc, phase flip on wrap, failure status preserved, non-zero reserved bytes rejected, and a window bounds check that overflow cannot defeat	R-4.11-001, R-9.5-002, R-9.5-005..010

[R-12.3-002] The suite SHALL build with -Wall -Wextra -Wpedantic -Werror.

13 Build System

```
cmake_minimum_required(VERSION 3.20)
# The document, uaf_config.h and this file MUST carry the same version. Each
# 2.1.x revision has changed a public signature, so they are not interchangeable
# and a mismatch would be silent.
project(uaf VERSION 2.1.4 LANGUAGES C)

set(CMAKE_C_STANDARD 11)
set(CMAKE_C_STANDARD_REQUIRED ON)
set(CMAKE_POSITION_INDEPENDENT_CODE ON)

option(UAF_BUILD_NVIDIA      "Build NVIDIA backend (UAF-N)"      OFF)
option(UAF_BUILD_SNAPDRAGON "Build Snapdragon backend (UAF-S)" OFF)
option(UAF_BUILD_DRAGONFLY  "Build Dragonfly backend (UAF-D)"  OFF)
option(UAF_BUILD_TESTS      "Build and register the conformance suite" ON)

add_library(uaf_core SHARED
    src/core/uaf_init.c      src/core/uaf_qp.c      src/core/uaf_cq.c
    src/core/uaf_mr.c        src/core/uaf_conn.c     src/core/uaf_storage.c
    src/core/uaf_wire.c      src/core/uaf_crc32c.c    src/core/uaf_ring.c
    src/core/uaf_rkey.c      src/core/uaf_dst.c      src/core/uaf_qp_state.c
)
target_include_directories(uaf_core PUBLIC include)

# A stable ABI needs a version on the shared object. v2.0 declared neither a
# SOVERSION nor an install rule for a library whose premise was ABI stability.
# SOVERSION is major.minor rather than major while the specification is
# pre-approval: v2.1.1 changed the poll signature and v2.1.2 changed
# uaf_create_qp, so a bare major number would claim an ABI these revisions do
# not share. v2.1.3 changed the connection record, a wire change rather than an
# ABI change, and v2.1.4 changes neither. It becomes major-only at the
# first approved release.
set_target_properties(uaf_core PROPERTIES
    VERSION    ${PROJECT_VERSION}
    SOVERSION  ${PROJECT_VERSION_MAJOR}.${PROJECT_VERSION_MINOR})

# One backend per process ([R-1.2-003]); each exports only uaf_pal_get_ops().
if(UAF_BUILD_NVIDIA)
    find_package(PkgConfig REQUIRED)
    pkg_check_modules(IBVERBS REQUIRED libibverbs)
    add_library(uaf_n SHARED src/pal/nvidia/uaf_pal_nvidia.c)
    target_link_libraries(uaf_n PRIVATE uaf_core ${IBVERBS_LIBRARIES} cufile)
endif()
if(UAF_BUILD_SNAPDRAGON)
    add_library(uaf_s SHARED src/pal/snapdragon/uaf_pal_snapdragon.c)
    target_link_libraries(uaf_s PRIVATE uaf_core uring)
endif()
if(UAF_BUILD_DRAGONFLY)
    add_library(uaf_d SHARED src/pal/dragonfly/uaf_pal_dragonfly.c)
    target_link_libraries(uaf_d PRIVATE uaf_core)
endif()

if(UAF_BUILD_TESTS)
    enable_testing()
    add_subdirectory(conformance)
endif()
```

```
include(GNUInstallDirs)
install(TARGETS uaf_core LIBRARY DESTINATION ${CMAKE_INSTALL_LIBDIR})
install(DIRECTORY include/ DESTINATION ${CMAKE_INSTALL_INCLUDEDIR}/uaf)
```

14 Appendices

14.1 Appendix A: Public header (uaf_core.h)

NORMATIVE.

```
#ifndef UAF_CORE_H
#define UAF_CORE_H

#include <stdint.h>
#include <stddef.h>
#include "uaf_config.h"
#include "uaf_types.h"

#ifdef __cplusplus
extern "C" {
#endif

/* Control functions return UAF_OK or a negative enum uaf_error.
 * Poll functions return a non-negative count or a negative error. */

int uaf_init(void);
int uaf_fini(void);
const char *uaf_version(void);
const char *uaf_backend_name(void);
uint32_t uaf_abi_version(void);

int uaf_get_device_list(uaf_device_t ***devs, int *count);
int uaf_free_device_list(uaf_device_t **devs, int count);
int uaf_query_device(uaf_device_t *dev, struct uaf_device_attr *attr);
int uaf_open_device(uaf_device_t *dev, uaf_domain_t **pd);
int uaf_close_device(uaf_device_t *dev);
int uaf_close_domain(uaf_domain_t *pd);

int uaf_reg_mr(uaf_domain_t *pd, void *addr, size_t len,
               uint64_t flags, uaf_mr_t **mr);
int uaf_alloc_mr(uaf_domain_t *pd, size_t len, uint64_t flags, uaf_mr_t **mr);
int uaf_dereg_mr(uaf_mr_t *mr);
int uaf_free_mr(uaf_mr_t *mr);

int uaf_create_cq(uaf_domain_t *pd, int cqe, uaf_cq_t **cq);
int uaf_destroy_cq(uaf_cq_t *cq);
int uaf_poll_cq(uaf_cq_t *cq, int max, struct uaf_wc *wc);

int uaf_create_qp(uaf_domain_t *pd, uaf_cq_t *scq, uaf_cq_t *rcq,
                  struct uaf_qp_init_attr *init, uaf_qp_t **qp);
int uaf_modify_qp(uaf_qp_t *qp, enum uaf_qp_state state,
                  struct uaf_qp_attr *attr, uint32_t attr_mask);
int uaf_destroy_qp(uaf_qp_t *qp);
int uaf_post_send(uaf_qp_t *qp, struct uaf_wr *wr);
int uaf_post_recv(uaf_qp_t *qp, struct uaf_wr *wr);
```

```

int uaf_get_conn_info(uaf_qp_t *qp, struct uaf_conn_info *local);
int uaf_connect(uaf_qp_t *qp, const struct uaf_conn_info *remote);
int uaf_disconnect(uaf_qp_t *qp);
int uaf_conn_info_serialize(const struct uaf_conn_info *ci, uint8_t out[64]);
int uaf_conn_info_deserialize(const uint8_t in[64], struct uaf_conn_info *ci);

int uaf_map_peer(uaf_domain_t *pd, const struct uaf_peer_info *peer,
                uaf_mr_t **mr);
int uaf_unmap_peer(uaf_mr_t *mr);

int uaf_storage_open(uaf_domain_t *pd, const char *path,
                    uaf_sq_t **sq, uaf_cr_t **cr);
int uaf_storage_submit(uaf_sq_t *sq, struct uaf_storage_cmd *cmd);
/* Fills the HOST completion, whose wr_id is the caller's full 64-bit value.
 * struct uaf_storage_cqe is the device ring image and is not returned here. */
int uaf_storage_poll(uaf_cr_t *cr, int max, struct uaf_storage_wc *wc);
int uaf_storage_close(uaf_sq_t *sq, uaf_cr_t *cr);
int uaf_storage_reset(uaf_sq_t *sq, uaf_cr_t *cr);

const char *uaf_strerror(int e);
void uaf_set_log_level(int level);
void uaf_set_log_callback(void (*cb)(int level, const char *msg));

#ifdef __cplusplus
}
#endif
#endif /* UAF_CORE_H */

```

14.2 Appendix B: Reference application (rdma_write_demo.c)

INFORMATIVE, but every return value is checked and the connection is established. v2.0's Appendix B never called `uaf_connect()`, wrote to remote address 0 with rkey 0 from a zeroed `uaf_conn_info`, then polled forever for a completion that could not arrive, and ignored twelve return values.

```

#include "uaf_core.h"
#include <stdio.h>
#include <string.h>

#define CK(expr) do { int _rc = (expr); if (_rc != UAF_OK) { \
    fprintf(stderr, "%s failed: %s\n", #expr, uaf_strerror(_rc)); \
    return 1; } } while (0)

/* Exchanges 64 serialised bytes with the peer; implemented by the caller
 * over TCP, a job launcher, or any out-of-band channel. */
extern int exchange_conn(const uint8_t out[64], uint8_t in[64]);

int main(void)
{
    CK(uaf_init());

    uaf_device_t **devs = NULL;
    int n = 0;
    CK(uaf_get_device_list(&devs, &n));
    if (n == 0) { fprintf(stderr, "no UAF device\n"); return 1; }
}

```

```

struct uaf_device_attr da = { .struct_size = sizeof(da) };
CK(uaf_query_device(devs[0], &da));
if (!(da.caps & UAF_CAP_ATOMICS))
    printf("note: atomics unavailable on this backend\n");

uaf_domain_t *pd = NULL;
CK(uaf_open_device(devs[0], &pd));

uaf_cq_t *cq = NULL;
CK(uaf_create_cq(pd, 64, &cq));

struct uaf_qp_init_attr init = {
    .struct_size = sizeof(init),
    .qp_type = UAF_QPT_RC,
    .path = UAF_PATH_UDP, /* [R-4.5-005]; 0 is not a value */
    .max_send_wr = 64, /* powers of two */
    .max_recv_wr = 64,
    .max_send_sge = 1,
    .max_recv_sge = 1,
    .max_inline_data = da.max_inline_data,
    .sq_sig_all = 0,
};
uaf_qp_t *qp = NULL;
CK(uaf_create_qp(pd, cq, cq, &init, &qp));

/* Always-portable buffer acquisition: the library owns the allocation,
 * so this works whether or not UAF_CAP_REG_ANY_MEM is set. */
uaf_mr_t *mr = NULL;
CK(uaf_alloc_mr(pd, 4096,
    UAF_MR_LOCAL_WRITE | UAF_MR_REMOTE_WRITE |
    UAF_MR_REMOTE_READ, &mr));
memset(mr->addr, 0, mr->length);
snprintf((char *)mr->addr, mr->length, "Hello, UAF v2.1.4!");
const uint32_t payload_len = (uint32_t)strlen((char *)mr->addr) + 1u;

/* Exchange connection information, then connect. */
struct uaf_conn_info local = { .struct_size = sizeof(local) };
CK(uaf_get_conn_info(qp, &local));
local.rkey = mr->rkey;
local.remote_va = (uint64_t)(uintptr_t)mr->addr;
local.path = init.path; /* MUST match the peer, [R-5.5-012] */

uint8_t tx[64], rx[64];
CK(uaf_conn_info_serialize(&local, tx));
if (exchange_conn(tx, rx) != 0) { fprintf(stderr, "exchange failed\n"); return 1; }

struct uaf_conn_info remote;
CK(uaf_conn_info_deserialize(rx, &remote));
CK(uaf_connect(qp, &remote)); /* drives INIT -> RTR -> RTS */

struct uaf_sge sge = {
    .addr = (uint64_t)(uintptr_t)mr->addr,
    .length = payload_len, /* not the whole 4 KiB region */
    .lkey = mr->lkey,
};
struct uaf_wr wr = {
    .wr_id = 1,
    .opcode = UAF_WR_RDMA_WRITE,

```

```

        .num_sge      = 1,
        .sge          = &sge,
        .send_flags   = UAF_SEND_SINGALED,
        .remote_addr  = remote.remote_va,
        .rkey         = remote.rkey,
    };
    CK(uaf_post_send(qp, &wr));

    /* Poll returns a COUNT: 0 means empty, negative means failure. */
    struct uaf_wc wc;
    for (;;) {
        int got = uaf_poll_cq(cq, 1, &wc);
        if (got < 0) {
            fprintf(stderr, "poll_cq: %s\n", uaf_strerror(got));
            return 1;
        }
        if (got == 1) break;
    }
    if (wc.status != UAF_OK) {
        fprintf(stderr, "write failed: %s\n", uaf_strerror(wc.status));
        return 1;
    }
    printf("write completed: %u bytes\n", wc.byte_len);

    CK(uaf_disconnect(qp));
    CK(uaf_free_mr(mr));
    CK(uaf_destroy_qp(qp));
    CK(uaf_destroy_cq(cq));
    CK(uaf_close_domain(pd));          /* absent from the v2.0 ABI */
    CK(uaf_close_device(devs[0]));
    CK(uaf_free_device_list(devs, n));
    CK(uaf_fini());
    return 0;
}

```

14.3 Appendix C: Requirement index

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[R-1.2-003]	Operating environment
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[R-9.4-001]	9.4 Device-visible ordering
[R-9.4-002]	9.4 Device-visible ordering
[R-9.4-003]	9.4 Device-visible ordering
[R-9.5-001]	9.5 Intra-host binding and the ring consumer
[R-9.5-002]	9.5 Intra-host binding and the ring consumer
[R-9.5-003]	9.5 Intra-host binding and the ring consumer
[R-9.5-004]	9.5 Intra-host binding and the ring consumer
[R-9.5-005]	9.5 Intra-host binding and the ring consumer
[R-9.5-006]	9.5 Intra-host binding and the ring consumer
[R-9.5-007]	9.5 Intra-host binding and the ring consumer
[R-9.5-008]	9.5 Intra-host binding and the ring consumer
[R-9.5-009]	9.5 Intra-host binding and the ring consumer
[R-9.5-010]	9.5 Intra-host binding and the ring consumer
[R-10.1-001]	10.1 Queue-pair state machine
[R-10.1-002]	10.1 Queue-pair state machine
[R-10.1-003]	10.1 Queue-pair state machine
[R-10.1-004]	10.1 Queue-pair state machine
[R-10.1-005]	10.1 Queue-pair state machine
[R-10.1-008]	10.1 Queue-pair state machine
[R-10.2-001]	10.2 Key encoding
[R-10.2-002]	10.2 Key encoding
[R-10.2-003]	10.2 Key encoding
[R-10.3-001]	10.3 Ring discipline
[R-10.3-002]	10.3 Ring discipline
[R-10.3-003]	10.3 Ring discipline
[R-10.3-004]	10.3 Ring discipline
[R-10.3-005]	10.3 Ring discipline
[R-11.1-001]	11.1 Threat model and mitigations
[R-11.1-002]	11.1 Threat model and mitigations
[R-11.1-003]	11.1 Threat model and mitigations
[R-11.1-004]	11.1 Threat model and mitigations
[R-11.1-005]	11.1 Threat model and mitigations
[R-11.2-001]	11.2 Key management
[R-11.2-002]	11.2 Key management
[R-11.2-003]	11.2 Key management
[R-11.2-004]	11.2 Key management
[R-11.2-005]	11.2 Key management
[R-11.2-006]	11.2 Key management
[R-11.2-007]	11.2 Key management
[R-11.2-008]	11.2 Key management
[R-12.1-001]	12.1 Measurement method
[R-12.1-002]	12.1 Measurement method
[R-12.2-001]	12.2 Targets
[R-12.2-002]	12.2.2 RMT bandwidth (unidirectional goodput, 1 MiB messages)

Requirement	Defined in
[R-12.2-003]	12.2.3 DST throughput (4 KiB random read, QD 128)
[R-12.2-004]	12.2.2 RMT bandwidth (unidirectional goodput, 1 MiB messages)
[R-12.3-001]	12.3 Conformance suite
[R-12.3-002]	12.3 Conformance suite

Total: 192 numbered requirements, each defined exactly once and each cited reference resolved.

14.4 Appendix D: Open issues

Every issue must be closed or accepted before this document may be marked approved, per [R-1.6-001]. Four are closed on the substance (OI-2, OI-4, OI-5, OI-6) and four are accepted as deliberate limits of the 2.1.x line (OI-1, OI-3, OI-7, OI-8). OI-7 is the only one that needs silicon.

ID	Issue	Disposition sought
OI-1	No asynchronous event channel: an application learns of link-down or a queue-pair fatal only by polling.	Accepted for 2.1.x: polling is sufficient for every required path. <code>uaf_get_async_event()</code> is scheduled for v2.2.
OI-2	DOCA GPUNetIO is unmapped.	Closed in v2.1.2: answered by [R-3.2-002]. UAF-N is Profile A; Profile B on that backend is OPTIONAL and, if declared, cannot claim <code>UAF_CAP_HW_CC</code> . No further binding is owed.
OI-3	Interrupt-driven completion (<code>UAF_CAP_CQ_EVENTS</code>) is declared but has no API.	Accepted for 2.1.x: the bit lets a backend advertise the hardware capability; the API is scheduled for v2.2. Polling costs a core, a known cost on Snapdragon.
OI-4	<code>UAF_QPT_UD</code> was declared and unspecified.	Closed in v2.1.1: the enumerator is withdrawn. A datagram type may be added by a later revision with its own specification.
OI-5	Multi-producer posting (<code>UAF_CAP_MT_POST</code>) was referenced and had no defined ring discipline.	Closed in v2.1.1: withdrawn. Rings are strictly single-producer, single-consumer ([R-4.9-001]); bit 9 of the capability word is reserved.
OI-6	The CM pre-shared fabric key had no distribution mechanism.	Closed in v2.1.1: declared an external dependency; see [R-11.2-007].
OI-7	Performance targets are unvalidated on silicon.	Accepted as targets: [R-12.1-001] already labels every figure a target and requires a conformance claim to report measurements. This is the one issue that needs hardware.

ID	Issue	Disposition sought
OI-8	A Windows binding for UAF-S (DirectStorage) is deferred.	Accepted: Windows is out of scope by [R-1.2-001] and belongs to UAF-SPEC-002. Nothing here depends on it.

14.5 Appendix E: Disposition of the v2.1.3 review

The review of v2.1.3 scored it 9.3 / 10, judged every v2.1.2 sign-off item closed, decoded the Section 5.8 connection-record vector independently and confirmed it, and recommended signing Section 1.6. It listed six items it would still edit without holding the signature. All six are closed here; the signature block remains empty, because only the named reviewers can fill it.

Finding	Disposition
[R-9.5-002]’s window check is stated as <code>remote_addr + L > cxl_size</code> , and a <code>uint64_t</code> addition wraps	Restated as <code>remote_addr > cxl_size L > cxl_size - remote_addr</code> . [R-9.5-010] makes the overflow-safe form a general requirement and gives the single helper used throughout; cases C-5 and C-12 drive the wrapping case
[R-9.5-004] cites the [R-11.2-004] throttle on a path with no packets and no source address	The throttles are excluded explicitly and the bounds check is named as the control. An invalid rkey on that path is a local programming error
[R-11.2-008]’s optional data-packet authenticator has no byte layout	Withdrawn. A revision that wants it must define the packet format first
[R-9.5-006] says the offset is “recorded at queue-pair creation” while [R-9.5-009] says both ends compute it	006 now defers to 009; the formula is the address
Section 6.7’s command-specific fallback row puts the mapping in the Name column	Realigned
[R-1.2-001] says “out of scope for v2.1”; Appendix B’s payload says v2.1	Both updated

14.6 Appendix F: Disposition of the v2.1.2 review

The review of v2.1.2 scored it 8.8 / 10, judged three of the four v2.1.1 seams closed, and found the fourth broken on the wire. All findings are closed in v2.1.3.

Finding	Disposition
The path discriminant does not survive the byte exchange. Byte 59 of the record is still reserved-and-zero, [R-5.5-004] has no CXL exemption, and Appendix B leaves path at 0 [R-5.5-012] is cited by C-9 and does not exist — Section 5.5 jumps from 011 to 013	path occupies record byte 59 and reserved shrinks to 60..63 ([R-5.5-012]); [R-5.5-004] exempts the CXL path; the Section 5.8 vector shows the byte; Appendix B sets it; case C-9 Written. The requirement index is now a build gate that fails on any cited-but-undefined requirement, a duplicate definition, or a requirement defined in a change-log section
The intra-host completion inherits [R-6.3-002] and [R-6.3-005] and skips [R-6.3-003], so a consumer starting at phase 0 accepts a zeroed ring	[R-9.5-007] now inherits the whole discipline, stating the initial expected phase of 1 and the wrap flip; case C-12 asserts the initial value
The Section 9.5 diagram shows publication as the advance of head	Redrawn: publication is the phase-byte store, and head is named as the agent’s own bookkeeping

Finding	Disposition
The completion-ring offset is “recorded at queue-pair creation” and is in no struct [R-9.5-002] names <code>remote_va</code> and <code>length</code> , neither a member of struct <code>uaf_wr</code> [R-9.5-003] reads as the CXL row alone	[R-9.5-009] gives the formula both ends compute Uses <code>remote_addr</code> and the sum of the SGE lengths Says base mask plus the CXL row, and spells the mask out
Section 6.7 uses three wrong NVMe code points	Corrected against Base 2.0d: PRP Offset Invalid 0x13, Attempted Write to Read Only Range command-specific 0x82, media 0x81 Unrecovered Read Error with the tags at 0x82-0x84. The table now carries the spec’s own names, and C-8 pins each value
The [R-11.2-004] commentary says unauthenticated failures never engage the throttle, without scope C-5’s requirement list stops at [R-11.2-006] Appendix C attributes fifteen requirements to change-log tables, and three sections share the title “Memory registration” The 2 s replay window assumes the clocks agree and never says so Section 1.6 points the v2.0 reviews at Appendix E; the <code>async-event</code> non-goal cites OI-3	Scoped: that is the rule where an authenticator exists, and [R-11.2-008] supplies the per-source counter where none does Reads R-11.2-001. .008 The index distinguishes a definition from a citation and carries the numbered section, e.g. 4.6 Memory registration [R-5.5-014] states the assumption, the synchronisation requirement and the failure mode Both corrected

14.7 Appendix G: Disposition of the v2.1.1 review

The review of v2.1.1 scored it 8.3 / 10, judged the eight v2.1 blockers closed in substance, and identified four seams in those fixes plus a set of stale references. All are closed in v2.1.2.

Finding	Disposition
S1 UAF-D’s RTR mask disagrees with Section 10.1, and nothing selects the intra-host path from the UDP path	enum <code>uaf_path</code> added; <code>uaf_qp_init_attr.path</code> and <code>uaf_conn_info.path</code> carry it; Section 10.1 gains three per-path RTR columns ([R-10.1-008]); [R-9.5-003] now defers to that table; cases C-4 and C-9
S2 The CM MAC input and the replay key are defined only for the 96-byte packets, and <code>CM_REP</code> carries a reason it has no field for	[R-5.5-006] defines the input per body size; [R-5.5-007] keys replay on the header <code>qp_id</code> ; the simultaneous-open reason goes in <code>imm_data</code> ; case C-11
S3 The 22-byte identity does not say which address or which port, and destination fields would rebuild the asymmetry	[R-5.5-013] specifies source address, source UDP port and own <code>qp_num</code> , with the GID and a zero port under Profile A; case C-11
S4 The intra-host completion has no layout and no chosen ordering mechanism	struct <code>uaf_rmt_cqe</code> (32 bytes, phase at offset 31) defined; [R-9.5-006]..[R-9.5-008] fix the ring, the phase-byte choice and the reserved-byte check; case C-12
[R-4.9-001] still referenced <code>UAF_CAP_MT_POST</code>	Rewritten: concurrency is undefined, with no capability that changes that
[R-1.2-001] still called <code>UAF_QPT_UD</code> optional and capability-gated	Rewritten: the enumerator is withdrawn
[R-5.4-002], [R-5.6-003] and the error table still named <code>UAF_OP_NAK</code>	All three use <code>NAK_INVALID</code> or <code>NAK_SEQ</code> ; the error table gains an RNR row and a CM-length row
The <code>imm_data</code> field description covered only the old NAK	Rewritten per opcode: immediate data, cumulative PSN, expected PSN, RNR delay, error code, CM reason

Finding	Disposition
With no link authenticator, the rkey counter never advances and a 32-bit key is scannable in seconds	[R-11.2-008]: throttle per source address in that configuration; queue-pair state still never changes; case C-5
UAF_ACK_REQ_INTERVAL, UAF_ACK_COALESCE_NS, UAF_RKEY_THROTTLE_NS absent from uaf_config.h	All three added to the listing in Section 3.4
Document, header and CMake versions disagreed	All three read 2.1.2; SOVERSION is major.minor while the line is pre-approval
The key samples register with queue-pair 0 and never bind	[R-11.2-003] installs the queue-pair component at connect; uaf_key_bind_qp() added; an unbound rkey never validates; cases C-5 and C-10
OI-1, OI-3, OI-7, OI-8 can be accepted; OI-2 is answered by [R-3.2-002]	Appendix D records four closures and four acceptances

14.8 Appendix H: Disposition of the v2.1 review

The review of v2.1 raised eight blockers and six corrections. All were closed in v2.1.1.

Finding	Disposition
B1 UAF-N cannot satisfy both [R-10.2-002] and [R-11.2-002]; verbs commonly returns one value for lkey and rkey	[R-10.2-001] scoped to backends owning their key space; [R-10.2-003] permits equal verbs keys; [R-11.2-001]..[R-11.2-003] and [R-11.2-005] scoped to the software table
B2 The 64-bit wr_id has nowhere to be returned; uaf_storage_poll writes a struct whose only identifier is 16 bits	struct uaf_storage_wc added ([R-6.3-006]); uaf_storage_poll() fills it; case C-7 checks a 64-bit round trip and an unknown cmd_id
B3 Simultaneous open compares different tuples on each side, so both peers may stay active or both go passive; RTU length contradicts [R-5.5-001]; UAF_CM_REASON_CLOSE has no value	Canonical 22-byte endpoint identity and a strict total order ([R-5.5-009]); per-opcode CM body length ([R-5.5-001]); enum uaf_cm_reason values assigned; case C-11
B4 NAK-RNR and ECN echo are behaviours with no encoding; no ACK schedule, so a Go-Back-N window may never advance	UAF_OP_NAK_SEQ, NAK_RNR, NAK_INVALID; UAF_WIRE_FLAG_ECN_ECHO; [R-5.6-008]..[R-5.6-010]; case C-3
B5 The rkey failure limit moves a queue pair to ERR, reopening the off-path kill that [R-5.7-001] closes	[R-11.2-004] now throttles, counts only authenticated failures, and never changes queue-pair state; case C-5 asserts unauthenticated failures are ignored
B6 uaf_conn_info, uaf_peer_info, enum uaf_send_flags, the UAF_MR_* bits, UAF_CM_REASON_* and UAF_CAP_MT_POST are referenced and undefined; struct_size applied unevenly	All defined in Section 4.3; UAF_CAP_MT_POST withdrawn; [R-4.1-003] states which structs carry struct_size and why
B7 Profile A is outside the atomic and completion rules, which live in a Profile-B-only section	New Section 4.11, profile-independent: [R-4.11-001]..[R-4.11-003]. The 0x88 encoding stays in Section 5.4
B8 UAF-D's intra-host path has no connection, no consumer and no NVMe status map	New Section 9.5: [R-9.5-001]..[R-9.5-005]. NVMe status mapped in Section 6.7
C1 The MAP_SYNC rationale is wrong: MAP_SHARED does not reliably fail	[R-9.1-002] rewritten against measurement. On Linux 6.18, MAP_SHARED MAP_SYNC succeeded on tmpfs with the flag dropped, and returned EOPNOTSUPP on another filesystem. The silent case is the dangerous one
C2 The phase rule has the producer's release and no consumer acquire	[R-6.3-005] requires an acquire load; the reference implementation uses an acquire atomic load
C3 buf_len - buf_offset underflows in the informative sample	Comparison reordered so the subtraction cannot wrap

Finding	Disposition
C4 nv_get_device_list leaks a non-NULL empty device list	Sample frees the list on the empty path
C5 4 KiB at 128 GB/s is 32 ns, not 34 ns	Corrected
C6 The CXL goodput target should be stated against usable, not raw, line rate	[R-12.2-004] added; raw, usable band and target all stated; target reduced to 900 Gbps
OI-4, OI-5, OI-6 are specification holes rather than silicon follow-ups	All three closed in v2.1.1; see Appendix D

14.9 Appendix I: Disposition of the v2.0 reviews

Both reviews are dated 2026-09-29. “R1” is the first, “R2” the second.

Finding	Source	Disposition
One wire format cannot describe three transports	R1, R2	Fixed: Section 3.2, two declared profiles; principle 5 rewritten
poll_cq return convention undefined	R1, R2	Fixed: [R-4.7-001]..[R-4.7-004], case C-7
No message framing; multi-packet messages inexpressible	R1	Fixed: Section 5.3, cases C-3 and C-10
Atomics unimplementable; no response packet	R1, R2	Fixed: Section 5.4, case C-3
DST CQE has no phase bit	R1, R2	Fixed: Section 6.3, case C-7
MAP_SHARED \ MAP_SYNC cannot succeed	R1	Fixed: [R-9.1-002], [R-9.1-003]
Send ring wrapped on the wrong depth constant	R1, R2	Fixed: [R-9.2-001], case C-6
ibv_device use-after-free	R1	Fixed: [R-7.2-001]
nlb == 0 underflow	R1, R2	Fixed: [R-6.2-002], case C-8
UAF_WR_STORAGE_FLUSH issued as a write	R2	Fixed: [R-6.2-001], case C-8
UAF-S registration copies and relocates	R1, R2	Fixed: Section 4.6, [R-8.2-001], [R-8.2-002]
Key model contradicts the threat matrix	R1, R2	Fixed: Section 11.2, case C-5
dma_buf_fd in a connection record	R1, R2	Fixed: [R-5.5-005], case C-9
CRC32C miscast as an injection defence	R1, R2	Fixed: [R-11.1-003]
CM unspecified; no layout, auth, teardown or simultaneous open	R2	Fixed: Section 5.5
CM and ACK/NAK opcodes in no enum	R1, R2	Fixed: enum uaf_wire_opcode, case C-3
UAF_WR_RECV listed as a wire opcode	R2	Fixed: [R-5.3-002], case C-3
Reliability parameters inconsistent with the latency targets	R1, R2	Fixed: Section 5.6
DCQCN underspecified	R1, R2	Fixed: Section 5.9
create_qp has no depth, SGE or type; modify_qp has no mask	R2	Fixed: Section 4.5, case C-4
RTR demands GID and LID from backends without them	R2	Fixed: [R-10.1-002], case C-4
UAF_QPS_SQE unreachable	R1, R2	Fixed: [R-10.1-004], case C-4
path_mtu unit undefined	R1, R2	Fixed: [R-4.5-002]

Finding	Source	Disposition
Every fault flushes as UAF_ERR_QP_STATE	R2	Fixed: [R-10.1-005], case C-4
enum width and size_t in the ABI	R1, R2	Fixed: [R-4.1-001], case C-1
No struct size or version field	R2	Fixed: [R-4.1-003]
uaf_mr.pal_priv in the application ABI	R1, R2	Fixed: [R-4.1-005], case C-1
packed CQE drops alignment to 1	R1	Fixed: [R-6.3-001], case C-1
No uaf_query_device	R1, R2	Fixed: Section 4.8
No domain destructor	R1	Fixed: uaf_close_domain()
storage_close releases only the submission queue	R1, R2	Fixed: [R-6.4-002]
No storage_reset for the ERROR edge	R1	Fixed: [R-6.4-001]
Section 6.1 pseudo-SQE never used	R1, R2	Fixed: deleted; Section 6.2 is an NVMe mapping
cmd_id truncates wr_id	R1, R2	Fixed: Section 6.5, case C-8
Single PRP cannot describe a multi-page buffer	R2	Fixed: [R-6.2-003], case C-8
PAL linking unspecified; all three symbols in one header	R2	Fixed: Section 4.10
Threading unspecified	R1, R2	Fixed: Section 4.9
WR and SGE ownership at post time unspecified	R2	Fixed: [R-4.4-002]
SGE list cannot be expressed on the wire	R1, R2	Fixed: [R-4.4-004]
imm_data validity unsignalled	R1	Fixed: [R-4.4-001]
rnr_retry has no behaviour	R2	Fixed: [R-5.6-006]
Barrier class wrong for AArch64 MMIO	R1, R2	Fixed: Section 9.4
Submission-queue memory type unstated	R2	Fixed: [R-6.6-001]
UAF_D0ORBELL_SQ undefined	R2	Fixed: [R-9.2-003]
Device-DAX alignment unstated	R2	Fixed: [R-9.1-004]
"SQ" means two different queues	R2	Fixed: STQ in Section 2; Sections 9.2 and 9.3 disambiguated
remote_register_buf_attr truncates size_t	R2	Fixed: [R-8.2-004]
sd_post_send may submit twice	R2	Fixed: [R-8.3-001]
Non-coherent buffer never synchronised	R1, R2	Fixed: [R-8.2-003]
ibv_reg_mr failures all become UAF_ERR_NOMEM	R2	Fixed: [R-7.3-003]
UAF_MR_GPU ignored; no dma-buf path	R1, R2	Fixed: [R-7.3-001], [R-7.3-002]
cuFile path incomplete	R2	Fixed: [R-7.4-001]
MTU excludes the header	R1	Fixed: Section 3.4
CXL 3.1 paired with PCIe Gen7	R2	Fixed: [R-3.3-001]
UAF_ENABLE_CXL_CACHE off while CXL.cache advertised	R2	Fixed: [R-3.3-002]
CXL bandwidth double-counts the duplex link	R1, R2	Fixed: [R-12.2-002]
200 M IOPS and 0.3 μ s not credible	R1, R2	Fixed: [R-12.2-001], [R-12.2-003]
Performance table has no method	R1, R2	Fixed: Section 12.1

Finding	Source	Disposition
No conformance surface, no requirement IDs, no SHALL	R1, R2	Fixed: Section 1.4, Section 12.3, Appendix C
No approver or open-issues list	R2	Fixed: Section 1.6, Appendix D
Windows and Linux in one backend	R2	Fixed: [R-1.2-001], [R-8.1-002]
Non-goals and OS matrix absent	R2	Fixed: Section 1.2
Reference application non-functional	R1, R2	Fixed: Appendix B
32-bit PSN versus IB's 24-bit	R1	Fixed: [R-5.6-002], case C-3
Payload outside every integrity check	R1, R2	Fixed: data_crc32c, [R-5.2-002]
Bad magic or CRC could move a QP to ERR	R2	Fixed: [R-5.7-001], case C-10